

Why Rotating Fluids Make Polygons

Abstract

On the flat ($K = 0$) plane, the regular N -gon of point vortices is a constrained energy minimum for all $N \leq 7$ and unstable for $N \geq 8$ without a central vortex. This stability boundary is an invariant of the two-dimensional Laplacian: the logarithmic Green's function $G \propto -\ln r$ is the unique radially symmetric planar interaction for which the Havelock eigenvalue $\lambda_m = (N - 1) - m(N - m)/2$ is scale-independent, and $N = 7$ is the threshold where the leading eigenvalue $\lambda_3 = 0$ (the linear restoring force vanishes). A quartic calculation on the constraint surface confirms that the $N = 7$ ring is nonlinearly stable: the quartic coefficient $\alpha_0 = 45/14 \approx 3.21 > 0$ (in the normal form $\mathcal{H} = (\alpha_0/4)\varepsilon^4$) supplies a restoring force where the linear one is absent.

For $N \geq 8$ the stability boundary requires a central vortex of strength $\kappa_0/\kappa \geq \kappa_{\text{crit}}(N)$. The sharpest prediction is at $N = 8$: $\kappa_{\text{crit}}(8) = 1/2$. In a Bose–Einstein condensate, where circulation is quantized, this is an integer-valued threshold: a ring of eight unit-winding vortices is stable if and only if the central vortex carries winding number $q_0 \geq 1$.

Jupiter and Saturn are instances of 2D polygon-formation at different dynamical scales, connected by a shared geometric substrate. Jupiter's north polar octagon ($N = 8$) is a vortex-dominated ring; the ring energy H_N is strictly increasing in N (proven), so $N = 8$ is the unique maximum-energy configuration within the Thomson-stable set at $\kappa_0/\kappa \geq 1/2$. Selection of this maximum requires dynamics that preferentially populates higher-energy states—Onsager negative-temperature equilibrium being the standard mechanism, though not the only one consistent with the prediction. Saturn's hexagon ($N = 6$) is a wave-dominated meander selected by Rossby stationarity of the quasi-geostrophic equation. The two mechanisms are physically distinct: the Jupiter argument uses the point vortex Hessian directly; the Saturn argument uses the QGPV eigenvalue problem. Both are governed by the 2D Laplacian—but this shared feature reflects two-dimensionality itself, not a deeper algebraic coincidence. The unification is structural (same Green's function, same stability threshold $N = 7$) rather than mechanistic (the paths to the polygon differ).

On the hyperbolic plane $\mathbf{H}^2$ with curvature $K = -1/a^2$, the stability boundary rises monotonically: $N_{\text{crit}} = 12$ at ring-to-curvature ratio $R/a = 1$. A 12-vortex ring stable without a central vortex is impossible in flat geometry; this is a falsifiable prediction for vortex dynamics on hyperbolic lattices now realizable in circuit QED [Kollár et al., 2019]. The $7 \rightarrow 8$ transition occurs at the unique $\xi^* \in (0, 1)$ satisfying $7(1 + \xi^2)/(1 - \xi^2)^2 = 8$, reducing to $\xi^2 - 16\xi + 1 = 0$ with positive root $\xi^* = 8 - 3\sqrt{7}$. The threshold polynomial $\xi^2 - 16\xi + 1 = 0$ is *palindromic* (equal leading and trailing coefficients), forcing its roots to multiply to 1: ξ^* is automatically an algebraic unit ($\xi^* = \varepsilon^{-1}$, where $\varepsilon = 8 + 3\sqrt{7}$ is the fundamental unit of $\mathbb{Z}[\sqrt{7}]$). The palindromic structure is not accidental—it reflects the conformal inversion symmetry $\xi \leftrightarrow 1/\xi$ of the Poincaré disk (Proposition 2), connecting the algebraic number theory of

the threshold to the Möbius covariance of the logarithmic energy. The 7 in $\mathbb{Q}(\sqrt{7})$ appears because the threshold condition has discriminant $252 = 4 \cdot 9 \cdot 7$.

1 Introduction

These observations raise a question with two parts. The first—*how* does a specific polygon form?—has received considerable attention. The second—*why* should different physical mechanisms on different planets all produce the same geometric outcome?—has not.

In 1988, David Godfrey examined Voyager images of Saturn’s north pole and found a hexagon [Godfrey, 1988]—a persistent, regular, six-sided polygon inscribed in the jet stream at 76°N , stable across decades. Its most striking property is that it co-rotates with Saturn’s System III period ($10^{\text{h}} 39^{\text{m}} 22.4^{\text{s}}$), the rotation rate of the deep interior inferred from the magnetic field, rather than being advected by the $\sim 120\text{ m/s}$ eastward jet in which it sits [Godfrey, 1988, Sánchez-Lavega et al., 2014]. This phase-locking to the deep interior has been confirmed from Voyager through Cassini across more than 30 years and is the primary observational constraint on any proposed mechanism. Thirty years later, the Juno spacecraft revealed that Jupiter’s poles harbor their own polygonal arrangements: eight cyclones in an octagonal ring at the north pole, five in a pentagonal ring at the south [Adriani et al., 2018]. Numerical simulations can reproduce these structures [Morales-Juberías et al., 2011, Gavriel and Kaspi, 2021], and vortex crystals emerge spontaneously in 2D turbulence simulations on polar caps [Siegelman et al., 2022].

We answer the second question, subject to one physical assumption (Onsager negative-temperature statistics at the polar scale; see Section 5 for a detailed assessment). The mechanism operates at two scales:

1. **Macro scale: cluster formation.** The Kraichnan (1967) inverse energy cascade drives 2D turbulence toward large-scale coherent structures. In the Onsager (1949) framework, this produces a negative-temperature state where same-sign vorticity concentrates into N coherent clusters [Bouchet and Venaille, 2012, Siegelman et al., 2022]. This step explains why discrete vortex clusters exist; it does not determine their spatial arrangement.
2. **Meso scale: polygon arrangement.** Given N vortex clusters on a ring of fixed angular momentum, the regular N -gon *minimizes* the interaction energy on the conserved-quantity surface (Theorem 5). By the energy-Casimir method, this makes the N -gon a nonlinearly stable equilibrium for $N \leq 7$ without a central vortex, or for any N with sufficient central vortex strength. The polygon is selected by dynamical stability, not by statistical mechanics.

The meso-scale proof (Theorem 5) applies to discrete point vortices and is directly relevant to Jupiter’s cyclone rings. Saturn’s hexagon is a continuous jet meander, not a discrete vortex ring; its wavenumber $N = 6$ is selected instead by Rossby wave stationarity of the QGPV equation (Proposition 8). Both mechanisms share the logarithmic Green’s function as a structural foundation, but the mathematical arguments differ.

Which polygon forms is determined by planet-specific selection criteria: $N = 6$ by Rossby stationarity (Saturn), $N = 8$ by the Thomson–Onsager maximum-stability bound (Jupiter north). The observed $N = 5$ at Jupiter’s south pole is consistent with Thomson stability ($\kappa_{\text{crit}}(5) = 0$, so no central vortex is required) but is not uniquely predicted by the

present framework; the south-polar packing geometry provides the additional constraint (Section 5.2).

Beyond the solar system, the framework makes its most physically distinctive prediction on the hyperbolic plane: a ring of $N = 12$ vortices at $R/a = 1$ is stable *without* a central vortex—an impossibility in flat geometry, where any $N \geq 8$ ring requires central stabilization. Recent implementations of hyperbolic lattice Hamiltonians in circuit QED [Kollár et al., 2019] and topoelectrical circuits [Lenggenhager et al., 2022] make this prediction experimentally accessible (Section 4).

2 The logarithmic interaction and its geometry

2.1 Point vortex dynamics on a 2D surface

On a two-dimensional surface, the stream function ψ satisfies the quasi-geostrophic potential vorticity (QGPV) equation

$$\frac{Dq}{Dt} = 0, \quad q = \nabla^2 \psi + \beta y. \quad (1)$$

The Green's function of the 2D Laplacian is $G(r) = -\ln r/(2\pi)$, satisfying $G'(r) = -1/(2\pi r) < 0$ for all $r > 0$.

For N point vortices with equal circulations κ on the collision-free configuration space $\{|z_j - z_k| > 0 \text{ for all } j \neq k\}$, where H is smooth and $\mathbb{Z}_N$ -invariant, the interaction energy is

$$H = -\sum_{j < k} \ln |z_j - z_k|, \quad (2)$$

where we absorb $\kappa^2/(4\pi)$ into the normalization. By the equivariant critical point theorem, the regular N -gon—a fixed point of $\mathbb{Z}_N$ —is a critical point of H .

Point vortex dynamics conserves three quantities: the energy H , the angular impulse $L = \sum |z_k|^2$, and the linear impulse $\mathbf{P} = \sum z_k$. The physically accessible configuration space is the $(2N - 3)$ -dimensional surface $\{L = NR^2, \mathbf{P} = 0\}$.

2.2 The sign rule (proven)

Theorem 1 (Sign rule for general interactions). *For $N \geq 3$ vortices on a ring with general pairwise interaction $h(r)$, the energy along the centered spiral satisfies*

$$H''(0) = \sum_{m=1}^{N-1} \left[(d_m^2 h''(d_m) + d_m h'(d_m)) S_m + \frac{h'(d_m) m^2 (N - m)}{N^2 d_m} \right], \quad (3)$$

where $d_m = 2R \sin(\pi m/N)$ and $S_m = \sum_{k=0}^{N-m-1} [(2k + m - (N-1))/(2N)]^2 \geq 0$.

If $h'(r) < 0$ and $(rh'(r))' \leq 0$ for all $r > 0$, then $H''(0) < 0$.

Proof. Step 1: Pairwise distance as a function of σ . For vortices j and k with separation $m = j - k > 0$, write the centered exponent as $(j - (N-1)/2)/N = (j' + k')/2 + m/(2N)$ where $j' = (j - (N-1)/2)/N$ and $k' = (k - (N-1)/2)/N$. Define the mean displacement $s = \sigma(j' + k')/2$ and half-separation $t = \sigma m/(2N)$. Then $z_j(\sigma) = R e^{s+t} e^{i\theta_j}$, $z_k(\sigma) = R e^{s-t} e^{i\theta_k}$ with $\theta_j - \theta_k = 2\pi m/N$, giving

$$D(\sigma) := |z_j - z_k|^2 = R^2 e^{2s} [e^{2t} + e^{-2t} - 2 \cos(2\pi m/N)] = R^2 e^{2s} [2 \cosh(2t) - 2 \cos(2\pi m/N)].$$

At $\sigma = 0$: $s = t = 0$, so $D(0) = 2R^2[1 - \cos(2\pi m/N)] = 4R^2 \sin^2(\pi m/N) = d_m^2$.

Step 2: First and second derivatives of D . Write $c_s = (j' + k')/2$ and $c_t = m/(2N)$, so $s = \sigma c_s$, $t = \sigma c_t$. Direct differentiation gives

$$\begin{aligned} D'(0) &= 4c_s(1 - \cos \phi), & \phi &= 2\pi m/N, \\ D''(0) &= 8c_s^2(1 - \cos \phi) + 8c_t^2. \end{aligned}$$

Step 3: Chain rule for $h(\sqrt{D})$. Write $d = \sqrt{D}$. Two applications of the chain rule give

$$\left. \frac{d^2}{d\sigma^2} h(d) \right|_{\sigma=0} = \underbrace{\frac{h''(d_m)}{4d_m^2}(D')^2 + \frac{h'(d_m)}{2d_m} \left[D'' - \frac{(D')^2}{2d_m^2} \right]}_{\text{evaluated at each pair } (j, k)}.$$

Substituting $D' = 4c_s d_m^2/2$ (since $1 - \cos \phi = d_m^2/(2R^2)$, and absorbing R into d_m) and collecting terms:

$$\left. \frac{d^2}{d\sigma^2} h(d) \right|_0 = [d_m^2 h''(d_m) + d_m h'(d_m)] c_s^2 + \frac{h'(d_m)}{d_m} c_t^2 \frac{d_m^2}{(1 - \cos \phi)}.$$

Step 4: Summation over pairs. For fixed separation m , sum over the $N - m$ pairs $(k, k+m)$ for $k = 0, \dots, N-m-1$. Define

$$S_m = \sum_{k=0}^{N-m-1} c_{s,k}^2, \quad \text{where } c_{s,k} = \frac{2k + m - (N-1)}{2N}.$$

Since $\sum_k c_{s,k} = 0$ by the centering, $S_m \geq 0$ with equality only if all $c_{s,k} = 0$ (which requires $N-m=1$, i.e. $m = N-1$). The second sum contributes $c_t^2 \cdot (N-m) = m^2(N-m)/(4N^2)$, and $d_m^2/(1 - \cos \phi) = 4R^2$. Summing over $m = 1, \dots, N-1$ yields (3).

Step 5: Sign determination. Under $h'(r) < 0$: the second term $h'(d_m) m^2(N-m)/(N^2 d_m) < 0$ for all $m = 1, \dots, N-1$ (including $m = N-1$, where $S_{N-1} = 0$ but the second term is still negative). Under $(rh'(r))' \leq 0$: the first coefficient $d_m^2 h''(d_m) + d_m h'(d_m) = d_m \cdot (rh')'|_{r=d_m} \leq 0$, so the first term is ≤ 0 since $S_m \geq 0$. (At $m = N-1$: $S_{N-1} = 0$ and the first term vanishes regardless of the sign condition; the inequality $H''(0) < 0$ follows from the strictly negative second terms alone.) Therefore $H''(0) < 0$. $\square$

Remark 1 (Borderline status of the logarithmic interaction). *The sufficient conditions of Theorem 1 partition interactions into three classes:*

- (i) $h' < 0$ and $(rh')' < 0$: both terms in (3) are strictly negative. Examples: $h = -r^\alpha$ ($\alpha > 0$), for which $(rh')' = -\alpha^2 r^{\alpha-1} < 0$.
- (ii) $h' < 0$ and $(rh')' = 0$: the first term vanishes and only the second contributes. For smooth h on $(0, \infty)$, this class contains a unique functional form, up to positive scaling and additive constant: $(rh')' = 0 \Rightarrow rh' = c \Rightarrow h' = c/r \Rightarrow h = c \ln r + b$; combined with $h' < 0$ gives $c < 0$, i.e. $h(r) = -a \ln r + b$ with $a > 0$.
- (iii) $h' < 0$ and $(rh')' > 0$ at some r : the sign of the first term is indefinite and $H''(0) < 0$ is not guaranteed.

The 2D Green's function $h(r) = -\ln r$ falls in class (ii)—the boundary between guaranteed and non-guaranteed negativity. In this precise sense, the logarithmic interaction produces the weakest $|H''(0)|$ among all interactions in classes (i) and (ii).

Remark 2 (The logarithm as the unique dilation-invariant interaction). Among all pairwise interactions $h : (0, \infty) \rightarrow \mathbb{R}$ with $h' < 0$, the logarithmic interaction $h(r) = -\ln r$ is the unique interaction invariant under spatial dilation up to an additive constant. Define the scale map $T_b[h](r) = h(br)$ ($b > 0$). Dilation-invariance means $h(br) = h(r) + c(b)$ for all r —that is, rescaling changes h only by an additive constant (irrelevant to the force $-h'$). This is a Cauchy functional equation whose continuous solutions with $h' < 0$ are exactly $h(r) = -a \ln r + b$, $a > 0$, confirming class (ii) as the unique dilation-invariant class. The N -body lift of this per-pair statement is Corollary 3: the Möbius covariance of H (Proposition 2), restricted to the dilation subgroup, gives $H(bz_1, \dots, bz_N) = H(z_1, \dots, z_N) - \frac{N(N-1)}{2} \ln b$, which is position-independent—the same fixed-point condition at the N -body level. This symmetry is not merely a geometric observation about the logarithm: it is the structural reason for the scale-independence of $N_{\text{crit}} = 7$. Because scaling changes H by an additive constant (Corollary 3), the Havelock eigenvalue $\lambda_m = (N-1) - m(N-m)/2$ —which determines whether the N -gon is a stable constrained minimum—is scale-independent, and the threshold $N = 7$ cannot drift under spatial rescaling. Prior treatments of Möbius covariance in vortex dynamics (e.g. Newton 2001) identify it as a conformal symmetry of the stream function; Corollary 3 shows it additionally implies that the stability threshold is dilation-invariant.

Under linearization, a perturbation $g(r) = r^\alpha$ transforms as $T_b[g](r) = b^\alpha r^\alpha$. The scaling exponent is b^α : perturbations with $\alpha < 0$ (decaying at large r) are suppressed under rescaling—the coefficient $b^\alpha \rightarrow 0$ as $b \rightarrow \infty$; perturbations with $\alpha > 0$ are amplified under rescaling—the coefficient grows. Notably, the class (i) interactions $h = -r^\beta$ ($\beta > 0$) are amplified by rescaling: they depart from the logarithm at large scales, becoming stronger confining potentials. Despite this, class (i) still satisfies the sign rule and yields polygon equilibria, so polygon formation is robust across both the dilation-invariant interaction and interactions amplified by rescaling.

The stable manifold of $h = -\ln r$ —interactions of the form $-\ln r$ plus corrections that decay at large r —arises naturally from compact vorticity distributions. For $|z| < \varepsilon \ll |w| = d$, expand $\ln |w - z| = \ln d - \text{Re}(z/w) - \frac{1}{2} \text{Re}(z^2/w^2) + O(\varepsilon^3/d^3)$. Integrating against vorticity $\omega(z) d^2z$ gives $\int \omega(z) \ln |w - z| d^2z = K \ln d - d^{-1} \text{Re}(e^{-i\theta} \int z \omega d^2z) + O(\varepsilon^2/d^2)$. When the vortex centroid lies at the origin the dipole term vanishes, leaving an effective pairwise interaction $H_{\text{eff}}(d) = -K^2 \ln d + O(\varepsilon^2/d^2)$ for separations $d \gg \varepsilon$. The quadrupole corrections are suppressed as $d/\varepsilon \rightarrow \infty$ and vanish in the far-field limit. In the Kraichnan inverse-energy-cascade picture Kraichnan [1967], vorticity concentrates into compact structures of growing scale $L(t)$; at each scale the dominant inter-blob interaction is the logarithm, with higher-multipole corrections suppressed by $(L/d)^2 \rightarrow 0$. The cascade therefore drives the system toward the dilation-invariant logarithmic interaction, where the sign-rule polygon equilibria apply. This provides a physical explanation for why the sign rule—proved for a specific functional form—describes systems whose vorticity is not literally point-like: at the relevant scale $d \gg L$, the effective interaction is the logarithm to high accuracy.

The Gaussian curvature K is a destabilizing perturbation when $K > 0$ and a stabilizing one when $K < 0$. On the sphere ($K > 0$) the deviation from the logarithmic Green's function lowers N_{crit} below 7; on the hyperbolic plane ($K < 0$) it adds a positive $O(|K|)$ correction $12|K|$ to the formerly-marginal λ_3 eigenvalue (21). The flat-plane case $K = 0$ is therefore the least stable point of the constant-curvature family with respect to polygon formation: $N_{\text{crit}} = 7$ is the minimum over all constant-curvature surfaces.

2.3 Möbius covariance of the logarithmic energy

Proposition 2 (Möbius covariance). *Under a conformal map $f: \mathbb{C} \rightarrow \mathbb{C}$, the logarithmic energy transforms as*

$$H(f(z_1), \dots, f(z_N)) = H(z_1, \dots, z_N) - \sum_{k=1}^N \frac{N-1}{2} \ln |f'(z_k)|.$$

Consequently, the point vortex equations of motion are Möbius-invariant: the single-site correction $\frac{N-1}{2} \ln |f'(z_k)|$ depends only on z_k , so $\partial/\partial \bar{z}_j$ for $j \neq k$ annihilates it, and the Hamiltonian gradient $\partial H/\partial \bar{z}_j$ is unchanged.

Proof. The identity $|f(z) - f(w)| = |f'(z)|^{1/2} |f'(w)|^{1/2} |z - w| \cdot (1 + O(|z - w|))$ is exact when f is Möbius (the error term vanishes), so $\ln |f(z_j) - f(z_k)| = \ln |z_j - z_k| + \frac{1}{2} \ln |f'(z_j)| + \frac{1}{2} \ln |f'(z_k)|$. Summing over all $\binom{N}{2}$ pairs, each $\ln |f'(z_k)|$ appears $N-1$ times. Since $\frac{1}{2} \ln |f'(z_k)|$ depends on z_k alone, $\partial[\sum_k \frac{N-1}{2} \ln |f'(z_k)|]/\partial \bar{z}_j = \frac{N-1}{2} \partial \ln |f'(z_j)|/\partial \bar{z}_j$, which is the single-site term in the j -th equation of motion. For Möbius f , f' is rational and $\ln |f'|$ is harmonic away from poles, so this term contributes a gauge-like shift to the Hamiltonian that does not couple distinct vortices. $\square$

Corollary 3 (Dilation subgroup = scale-invariant Hamiltonian). *Among all Möbius transformations, the dilation subgroup $\{f(z) = bz : b > 0\}$ is uniquely characterised by having constant derivative: $f'(z) = b$ for all z . Substituting into Proposition 2 gives a correction that is position-independent:*

$$H(bz_1, \dots, bz_N) = H(z_1, \dots, z_N) - \frac{N(N-1)}{2} \ln b. \quad (4)$$

This is the N -body form of the dilation-invariance condition of Remark 2: scaling changes H only by an additive constant, so the equations of motion $\dot{z}_j = \partial H/\partial z_j$ are dilation-invariant. For any other Möbius map f , the derivative $|f'(z_k)|$ varies between sites, producing a configuration-dependent correction that is not a pure constant; the Hamiltonian is not scale-invariant under general conformal maps, only under the one-parameter dilation subgroup.

Equation (4) is the N -body lift of the per-pair uniqueness result in Remark 2: the logarithmic energy is the unique N -body Hamiltonian that is dilation-invariant.

Remark 3 (From sign rule to physical stability). *Theorem 1 establishes $H''(0) < 0$ along the spiral direction—an unconstrained energy curvature. This does not by itself imply instability, because the spiral deformation changes the conserved angular impulse $L = \sum |z_k|^2$ at $O(\sigma^2)$. The physical dynamics is constrained to $\{L = \text{const}\}$, and the Lagrange multiplier $\mu_L < 0$ shifts the curvature by $-2\mu_L > 0$, converting the unconstrained saddle into a constrained minimum (Theorem 5 below). The sign rule therefore plays an indirect but essential role: it guarantees that the N -gon is a saddle of the unconstrained energy with a specific sign structure ($H''(0) < 0$ along the spiral, $H''(0) > 0$ along other directions), and the Lagrange shift converts this saddle into a minimum for $N \leq 7$. For $N \geq 8$, the negative curvature along the $m = \lfloor N/2 \rfloor$ Fourier mode overwhelms the Lagrange shift, and the constrained configuration is a saddle.*

3 Stability of vortex rings on the flat plane

3.1 The centered spiral deformation

Consider N vortices on a ring of radius R , deformed along the one-parameter family

$$z_k(\sigma) = R e^{\sigma(k-(N-1)/2)/N} e^{2\pi i k/N}, \quad k = 0, \dots, N-1. \quad (5)$$

At $\sigma = 0$: the regular N -gon. The centering ensures $H(\sigma) = H(-\sigma)$ by the symmetry $k \mapsto N-1-k$ (which sends $\sigma \rightarrow -\sigma$ while preserving H), so $H'(0) = 0$. The spiral deformation has a precise Möbius-group origin: the positions $\{z_k(\sigma)\}$ are the orbit of the seed $z_0 = R$ under the cyclic group generated by $z \mapsto \lambda z$, $\lambda = e^{\sigma/N + 2\pi i/N}$. At $\sigma = 0$ this generator is purely elliptic (rotation by $2\pi/N$, recovering the regular N -gon); for $\sigma \neq 0$ it is loxodromic ($|\lambda| \neq 1$) and the orbit traces the logarithmic spiral $r = R e^{(\sigma/2\pi)\theta}$ at angles $\theta = 2\pi k/N$. The spiral deformation is therefore the generator of the loxodromic one-parameter subgroup through the N -gon in the Möbius group's action on configuration space. $\mathbb{Z}_N$ -equivariant perturbations of the N -gon form a vector space; the spiral direction is singled out within this space as the unique direction tangent to a Möbius orbit (generated by a single transformation $z \mapsto \lambda z$ acting on a seed point and producing all N positions via the $\mathbb{Z}_N$ orbit). Other perturbation directions (individual Fourier modes, radial displacements) do not arise from a single Möbius seed and are analyzed through the full constrained Hessian in Theorem 5. The Möbius interpretation does not enter the stability proofs: the stability analysis uses the spiral only as a specific perturbation direction (the unique $\mathbb{Z}_N$ -equivariant direction that leaves the constraint surface at $O(\sigma^2)$, so that the unconstrained $H''(0) < 0$ does not contradict the constrained energy minimum). However, Proposition 2 does carry genuine mathematical content beyond geometric decoration. Restricted to the dilation subgroup $f(z) = bz$, where f' is the constant b , the single-site correction becomes position-independent and reduces to $H(bz_1, \dots, bz_N) = H(z_1, \dots, z_N) - \frac{N(N-1)}{2} \ln b$. This is the N -body statement that H is dilation-invariant—the logarithmic energy changes only by an additive constant under spatial rescaling (Corollary 3 above).

3.2 Energy curvature along the spiral

Theorem 4 (Energy curvature at the polygon). *For $N \geq 3$ vortices on a ring with $h(r) = -\ln r$, the energy along the centered spiral deformation satisfies $H'(0) = 0$ and*

$$H''(0) = -\frac{1}{4N^2} \sum_{m=1}^{N-1} \frac{(N-m)m^2}{\sin^2(\pi m/N)} < 0. \quad (6)$$

Proof. Part (a): $H'(0) = 0$ by the centering symmetry $k \mapsto N-1-k$. Part (b): Define $F(u, m) = \ln[1 - 2\cos(2\pi m/N) e^{um} + e^{2um}]$ with $u = \sigma/N$. Direct differentiation gives $F''(0, m) = m^2/[2\sin^2(\pi m/N)]$. There are $(N-m)$ pairs with separation m , and the chain rule $d/d\sigma = (1/N)d/du$ contributes $1/N^2$. This derivation uses the specific structure of $h = -\ln r$: the general formula (3) has two terms, but for the logarithmic interaction $(rh')' = (-1)' = 0$, so the S_m terms (first term in (3)) vanish identically. Only the second term survives, giving (6) as a special case of (3) with $A_m = 0$ and $B_m = h'(d_m)m^2(N-m)/(N^2d_m) = -m^2(N-m)/(N^2d_m^2)$. Substituting $d_m = 2\sin(\pi m/N)$ yields (6). Every factor in the sum is strictly positive, so $H''(0) < 0$. $\square$

3.3 The constrained energy minimum

Attribution note. The Havelock eigenvalue formula $\lambda_m = (N-1) - m(N-m)/2$ and the linear stability boundary $N \leq 7$ are due to Havelock [1931]. The new contribution of the theorem below is the reframing: the N -gon is a *constrained energy minimum* (not a saddle point or maximum), which resolves the apparent tension with Onsager's statistical-mechanics picture.

Theorem 5 (Constrained energy minimum). *The regular N -gon minimizes H on the conserved-quantity surface $\{L = NR^2, \mathbf{P} = 0\}$ if and only if $N \leq 7$. The Lagrangian Hessian*

$$\nabla^2 \mathcal{L} = \nabla^2 H - 2\mu_L I, \quad \mu_L = -\frac{N-1}{4}, \quad (7)$$

restricted to the constraint tangent space, is positive semi-definite for $N \leq 7$ and indefinite for $N \geq 8$.

Proof. The energy-Casimir functional is $\mathcal{H} = H + |\mu_L| L$, whose Hessian is (7). The Hessian $\nabla^2 H$ is a $2N \times 2N$ real symmetric matrix that commutes with the $\mathbb{Z}_N$ action $z_k \mapsto z_{k+1 \pmod N}$ (since H is invariant under cyclic permutation of equal-circulation vortices). By Schur's lemma for real representations of finite abelian groups, $\nabla^2 H$ block-diagonalizes in the discrete Fourier basis into 2×2 blocks, one per mode $m \in \{0, \dots, N-1\}$. For mode m , decompose the perturbation at vortex k into radial (δr_k) and tangential (δt_k) components, with Fourier amplitudes $\delta r_k = a \cos(2\pi mk/N)$, $\delta t_k = b \cos(2\pi mk/N)$. The 2×2 Hessian block for the pair (a, b) involves two sums over separations $p = 1, \dots, N-1$:

$$\begin{aligned} H_{rr}^{(m)} &= \sum_p \left[\frac{1 - \cos(2\pi pm/N)}{4 \sin^2(\pi p/N)} + \frac{1 + \cos(2\pi pm/N)}{4 \sin^2(\pi p/N)} \right] - 2 \sum_p \frac{1 - \cos(2\pi pm/N)}{4 \sin^2(\pi p/N)}, \\ H_{rt}^{(m)} &= \sum_p \frac{\sin(2\pi pm/N)}{4 \sin^2(\pi p/N)}, \\ H_{tt}^{(m)} &= -H_{rr}^{(m)}. \end{aligned}$$

The off-diagonal $H_{rt}^{(m)}$ vanishes identically by the anti-symmetry $\sin(2\pi pm/N) \rightarrow -\sin(2\pi(N-p)m/N)$ under $p \rightarrow N-p$, since $\sin^2(\pi p/N) = \sin^2(\pi(N-p)/N)$ (so the denominator is symmetric while the numerator is anti-symmetric; the sum over the full range $p = 1, \dots, N-1$ cancels in pairs). Therefore the block is diagonal: $\text{diag}(H_{rr}^{(m)}, -H_{rr}^{(m)})$, giving eigenvalues $\pm H_{rr}^{(m)} = \pm \sigma_m$.

The diagonal entry $H_{rr}^{(m)}$ simplifies using the Havelock identity (Appendix A): the self-sum $\sum_p 1/[4 \sin^2(\pi p/N)] = (N-1)/2$ (a special case of $T_0 + (N-1)/2$) and the cross-sum is $T_m = m(N-m)/2$. Thus

$$\sigma_m = H_{rr}^{(m)} = \frac{N-1}{2} - \frac{m(N-m)}{2}, \quad (8)$$

using

$$T_m := \sum_{p=1}^{N-1} \frac{1 - \cos(2\pi pm/N)}{4 \sin^2(\pi p/N)} = \frac{m(N-m)}{2} \quad (9)$$

(proven in Appendix A; cf. Havelock [1931], §3). The Lagrangian eigenvalues are thus

$$\lambda_m^+ = (N-1) - \frac{m(N-m)}{2}, \quad \lambda_m^- = \frac{m(N-m)}{2}.$$

Since $\lambda_m^- \geq 0$ always, positive semi-definiteness requires $\lambda_m^+ \geq 0$ for the modes tangent to the constraint surface. Three modes are removed: *Mode* $m = 0$ corresponds to uniform dilation $\delta z_k \propto z_k$, which changes $L = \sum |z_k|^2$ and is therefore not tangent to $\{L = NR^2\}$. *Modes* $m = 1$ and $m = N - 1$ generate infinitesimal translation ($\delta z_k = \text{const}$, changing $\mathbf{P} = \sum z_k$) and overall rotation ($\delta z_k = iz_k$, a Goldstone mode of the $SO(2)$ symmetry of H). Both have $\sigma_m = 0$, consistent with the continuous symmetry group; they lie in the kernel of $\nabla^2 H$ and produce the zero eigenvalues. The remaining modes $m = 2, \dots, N - 2$ are tangent to the constraint surface and carry the stability information. The minimum of λ_m^+ over m occurs at $m = \lfloor N/2 \rfloor$:

$$\min_m \lambda_m^+ = \begin{cases} (N-1)(7-N)/8 & N \text{ odd,} \\ (N-1) - N^2/8 & N \text{ even.} \end{cases}$$

Both expressions are > 0 for $N \leq 6$, equal to zero at $N = 7$, and < 0 for $N \geq 8$. Thus $N \leq 6$ is *strictly* stable (positive definite Lagrangian), $N = 7$ is *neutrally* stable (positive semi-definite, with a zero eigenvalue at $m = 3$), consistent with Havelock [1931], and $N \geq 8$ is unstable. (The $N = 7$ marginal case is further analyzed in Section 3.5.) $\square$

Corollary 6 (Constrained minimum versus unconstrained saddle). *For $N \leq 7$, the N -gon is a constrained energy minimum on $\{L = NR^2, \mathbf{P} = 0\}$, yet a saddle point of the unconstrained energy H . These two facts are consistent, and their reconciliation corrects a widespread description in the statistical-mechanics literature of the N -gon as an energy maximum.*

Proof. The unconstrained Hessian $\nabla^2 H$ has $H_{rr}^{(m)} > 0$ and $H_{tt}^{(m)} < 0$ at the N -gon: radial perturbations raise the energy while tangential perturbations lower it (the polygon is a saddle point of the unconstrained H , consistent with Onsager's picture of local maxima in the energy landscape for same-sign vortices). The Lagrange multiplier $\mu_L = -(N-1)/4$ shifts all eigenvalues of $\nabla^2 H$ by $+2|\mu_L| = (N-1)/2$, converting the saddle into a minimum on the constraint surface. The value of μ_L follows from the constrained stationarity condition: at vortex 0 (position $(R, 0)$), the pair $(0, k)$ contributes identically $-1/(2R)$ to $\partial H/\partial x_0$ regardless of k , because $(1 - \cos \theta_k)/[4 \sin^2(\theta_k/2)] = 1/2$ for all k —a consequence of the scale invariance of $h'(r)/r = -1/r^2$. Summing over $N-1$ pairs gives $\partial H/\partial x_0 = -(N-1)/(2R)$, and setting this equal to $\mu_L \cdot 2R$ yields $\mu_L = -(N-1)/(4R^2)$.

In particular, Theorem 4 ($H''(0) < 0$ along the spiral direction) does not contradict Theorem 5: the spiral deformation (5) leaves the conserved surface at $O(\sigma^2)$, so it is not a constrained perturbation. The negative spiral curvature is an artifact of the angular impulse changing along the spiral path. $\square$

3.4 Central vortex stabilization

When a central vortex of circulation κ_0 is present at the origin, the central interaction $-\kappa_0 \sum_k \ln |z_k|$ modifies both the Lagrange multiplier and the Hessian. Two effects contribute to the critical eigenvalue:

Lagrange multiplier shift. The stationarity condition gives $\mu_L = -(N-1)/4 - \kappa_0/2$ (for $R = 1$), so the shift $-2\mu_L = (N-1)/2 + \kappa_0$ increases by $+\kappa_0$ per unit.

Hessian projection. The central-vortex Hessian at vortex k is the traceless 2×2 matrix $(\kappa_0/R^2) \begin{bmatrix} \cos 2\theta_k & \sin 2\theta_k \\ \sin 2\theta_k & -\cos 2\theta_k \end{bmatrix}$, which has eigenvalue $+\kappa_0/R^2$ for radial perturbations $(\cos \theta_k, \sin \theta_k)$ and $-\kappa_0/R^2$ for tangential. For $N \geq 8$, the critical eigenvector of the

ring-only Lagrangian (the most negative eigenvalue direction, at mode $m = \lfloor N/2 \rfloor$) is *purely radial* at every vortex position. This requires the off-diagonal block $H_{rt}^{(m)}$ to vanish at the critical mode $m = \lfloor N/2 \rfloor$, which we now verify explicitly. From equation (8): $H_{rt}^{(m)} = \sum_{p=1}^{N-1} \sin(2\pi pm/N)/[4\sin^2(\pi p/N)]$. The substitution $p \rightarrow N-p$ sends $\sin(2\pi pm/N) \rightarrow \sin(2\pi(N-p)m/N) = -\sin(2\pi pm/N)$ (anti-symmetric) while $\sin^2(\pi p/N) \rightarrow \sin^2(\pi(N-p)/N) = \sin^2(\pi p/N)$ (symmetric). Therefore each pair $(p, N-p)$ cancels, and $H_{rt}^{(m)} = 0$ for *every* m —in particular for the critical mode $m = \lfloor N/2 \rfloor$. The Hessian block is diagonal at all modes: $\text{diag}(H_{rr}^{(m)}, -H_{rr}^{(m)})$, so radial and tangential perturbations decouple completely. This decoupling is specific to the logarithmic (harmonic) interaction, for which $H_{tt} = -H_{rr}$. Therefore $\langle v_{\text{crit}} | \nabla^2 H_{\text{center}} | v_{\text{crit}} \rangle = +\kappa_0$.

The total shift of the critical eigenvalue is $+2\kappa_0$ per unit κ_0 (one from the Lagrange multiplier, one from the Hessian projection). Setting the shifted eigenvalue to zero:

Proposition 7 (Critical central vortex strength). *For $N \geq 8$, the N -gon with central vortex is a constrained energy minimum if and only if $\kappa_0/\kappa \geq \kappa_{\text{crit}}(N)$, where*

$$\kappa_{\text{crit}}(N) = \begin{cases} (N^2 - 8N + 8)/16 & N \text{ even}, \\ (N-1)(N-7)/16 & N \text{ odd}. \end{cases} \quad (10)$$

Proof. By Theorem 5, the minimum constrained Lagrangian eigenvalue at $\kappa_0 = 0$ is $\lambda_{\min} = (N-1) - \lfloor N/2 \rfloor (N - \lfloor N/2 \rfloor)/2$. The central vortex shifts this eigenvalue by $+2\kappa_0$: $+\kappa_0$ from the Lagrange multiplier (since $-2\mu_L$ increases by κ_0) and $+\kappa_0$ from the central-vortex Hessian projection (since the critical eigenvector is purely radial, and the central-vortex Hessian has eigenvalue $+\kappa_0/R^2$ on radial perturbations). Setting $(N-1) - m(N-m)/2 + 2\kappa_0 = 0$ with $m = \lfloor N/2 \rfloor$ gives (10). $\square$

The first values are:

N	≤ 7	8	9	10	11	12	13	14
κ_{crit}	0	$\frac{1}{2}$	1	$\frac{7}{4}$	$\frac{5}{2}$	$\frac{7}{2}$	$\frac{9}{2}$	$\frac{23}{4}$

These agree with the classical Thomson dynamical stability analysis [Thomson, 1883, Havelock, 1931], confirming that energy and dynamical stability criteria are equivalent.

For $N = 6$, the exact critical ratio $\kappa_{\text{crit}}(6) = -1/4$ follows algebraically from the shifted-eigenvalue formula in the proof of Proposition 7. With $N = 6$ (even) and critical mode $m = \lfloor N/2 \rfloor = 3$, the Lagrangian eigenvalue with central vortex at strength $r = \kappa_0/\kappa$ is

$$\lambda_3 = (N-1) - \frac{m(N-m)}{2} + 2r = 5 - \frac{9}{2} + 2r = \frac{1}{2} + 2r.$$

Setting $\lambda_3 = 0$ gives $r = -1/4$. The closed form $\kappa_{\text{crit}} = (N^2 - 8N + 8)/16$ of Proposition 7 evaluates at $N = 6$ to $(36 - 48 + 8)/16 = -4/16 = -1/4$, consistent with this. (The Proposition is stated for $N \geq 8$; at $N = 6$ the formula returns $\kappa_{\text{crit}} < 0$, meaning destabilisation requires an anticyclonic central vortex with $r < 0$.)

The same threshold appears in the dynamical stability analysis. The 12×12 linearisation of the hexagonal ring with central vortex [Havelock, 1931] has a characteristic polynomial containing the low-frequency factor

$$9 + 36\kappa_0/\kappa + 16\pi^2\lambda^2.$$

The constant term satisfies $9 + 36r = 18\lambda_3$, so the dynamical condition $9 + 36r = 0$ is algebraically equivalent to $\lambda_3 = 0$, both giving $r = -1/4$. For $\kappa_0/\kappa > -1/4$ the hexagonal configuration is stable; for $\kappa_0/\kappa < -1/4$ an anticyclonic central vortex destabilises it.

The center-vs-ring geometry is essential: multi-ring configurations are generically less stable (Section 6.3).

3.5 The $N = 7$ bifurcation

At $N = 7$, the minimum constrained Lagrangian eigenvalue is zero (mode $m = 3$). To classify the bifurcation, we compute the quartic term of H along the $m = 3$ neutral direction on the constraint surface. The $\mathbb{Z}_7$ -equivariant normal form is $H = (\lambda/2)|z|^2 + (\alpha/4)|z|^4 + \dots$ where z is the complex amplitude of the $m = 3$ mode and $\lambda = 0$ at $N = 7$. Numerical evaluation (tracing H along the Newton-projected constraint surface) gives $\alpha > 0$. The neutral subspace of the constrained Hessian consists of two modes: the radial Fourier modes at $m = 3$ and $m = 4 = N - 3$ (together forming the 2-dimensional space of symmetry-breaking perturbations), plus the rotation mode (trivially neutral by the $\text{SO}(2)$ symmetry of H).

For both neutral modes, the *unconstrained* quartic d^4H/dt^4 (where t is the unit-eigenvector amplitude) takes the same value by $\mathbb{Z}_7$ symmetry. Each of the 21 vortex pairs contributes

$$\frac{6C^2}{A^2} - \frac{12CB^2}{A^3} + \frac{3B^4}{A^4},$$

where $A = |\Delta z|^2$, $B = 2\text{Re}(\Delta z \cdot \overline{\Delta u})$, $C = |\Delta u|^2$ are evaluated at the unit-ring N -gon ($|\Delta z|^2 = 4\sin^2(\pi p/N)$ for separation class p). Direct summation over all 21 pairs gives

$$\left. \frac{d^4H}{dt^4} \right|_{m=3} = \frac{153}{7}. \quad (11)$$

The quartic sum is a polynomial in coordinates of the N -gon (whose positions are 7th roots of unity) and is $\mathbb{Z}_7$ -invariant: simultaneously shifting all vortex labels by one maps the ring to itself and maps u to a rotated copy. By Galois theory, such an invariant sum lies in $\mathbb{Q}(\zeta_7)^{\mathbb{Z}_7} = \mathbb{Q}$, so the result is necessarily a rational number. The value $153/7$ is then identified by the following argument: the 21 pairs decompose into 6 separation classes $p = 1, \dots, 6$ with 7 pairs each, and the per-pair quartic (11) depends on p only through $A_p = 4\sin^2(\pi p/7)$ and the mode overlap $B_p = 2\text{Re}(\Delta z \cdot \overline{\Delta u})$, $C_p = |\Delta u|^2$. These are algebraic numbers in $\mathbb{Q}(\zeta_7)$; summing with exact arithmetic (implemented in `n7_bifurcation.py`) gives $153/7$, verified independently by 50-decimal-place floating-point evaluation. The Galois rationality proof and the direct algebraic summation together constitute an exact result, not a numerical conjecture. (The $\mathbb{Z}_7$ symmetry forces the degree-4 form on the 2-dimensional neutral subspace to be proportional to $|z|^4$, so $153/7$ holds for *any* unit vector in the $\{m=3, m=4\}$ subspace.)

The constraint-surface correction admits an exact closed form via the *log-scaling identity*. The Newton projection onto $\{L = N\}$ sends $\varepsilon \mapsto z(\varepsilon) = (z + \varepsilon u) \sqrt{N/L(\varepsilon)}$, where $L(\varepsilon) = N + \varepsilon^2$ exactly (since $\text{Re} \sum_k z_k \overline{u_k} = 0$ for the radial $m=3$ mode and $\|u\|^2 = 1$). Because the logarithmic energy is a sum of $-\ln|\cdot|$ over $N(N-1)/2$ pairs, the overall scale factor $s(\varepsilon) = \sqrt{N/(N + \varepsilon^2)}$ contributes an additive term:

$$H_{\text{constrained}}(\varepsilon) = H_{\text{unconstrained}}(\varepsilon) + \frac{N(N-1)}{4} \ln \left(1 + \frac{\varepsilon^2}{N} \right). \quad (12)$$

Expanding the logarithm: $\ln(1 + \varepsilon^2/N) = \varepsilon^2/N - \varepsilon^4/(2N^2) + O(\varepsilon^6)$, so the 4th derivative of the correction term at $\varepsilon = 0$ is

$$\frac{N(N-1)}{4} \cdot \left(-\frac{12}{N^2}\right) = -\frac{3(N-1)}{N}.$$

For $N = 7$ this is $-18/7$. Combined with the unconstrained value $153/7$, the constrained quartic is therefore

$$\left.\frac{d^4 H}{dt^4}\right|_{\text{constrained}} = \frac{153}{7} - \frac{18}{7} = \frac{135}{7}, \quad (13)$$

an exact rational result. Richardson extrapolation (three two-point pairs at $h \in \{0.05, 0.04, 0.03, 0.02\}$, symmetric fourth-difference formula $[2\delta H(2h) - 8\delta H(h)]/h^4 \rightarrow 24c_4$ with log-ratio cancellation) gives 19.2857 ± 0.0005 , numerically consistent with the exact value $135/7 = 19.285714\dots$

Kurakin and Yudovich [2002] established nonlinear stability of the heptagonal ring using Arnold’s energy-Casimir method, proving positivity of the quartic form by Dirichlet’s criterion. Their analysis does not yield the explicit coefficient $\alpha_0 = 45/14$; rather, it establishes $\alpha_0 > 0$ by a convexity argument. The present derivation gives the exact value via the constrained energy-minimum framework (Theorem 5): the log-scaling identity (12) yields the exact correction $-3(N-1)/N$, and the unconstrained value $153/7$ is established by the Galois rationality argument combined with direct algebraic summation. In the normal form $\mathcal{H} = (\alpha_0/4)|z|^4$, the exact constrained quartic gives

$$\alpha_0 = \frac{135/7}{6} = \frac{45}{14} \approx 3.214 > 0: \quad (14)$$

the energy-Casimir functional $\mathcal{H} = H + |\mu_L|L$ has a strict local minimum along the $m = 3$ direction at quartic order, $\mathcal{H} \geq \mathcal{H}_0 + (\alpha_0/4)\varepsilon^4$. By the Dirichlet stability criterion [Boatto and Cabral, 2003], the heptagonal configuration is nonlinearly stable. The quartic (rather than quadratic) confinement explains why heptagonal rings are stable but more easily perturbed than hexagonal ones.

4 $N = 7$ as a curvature invariant

Prior work and what is new. Boatto and Cabral [2003] computed the constrained Lagrangian eigenvalues for a latitudinal ring of point vortices on the non-rotating sphere at arbitrary colatitude, establishing that the stability boundary shifts with latitude. Lim et al. [2001] studied relative equilibria of vortex rings on S^2 within the Hamiltonian framework. The spherical Havelock eigenvalues are therefore not novel.

What the proof in §4.1 adds is an explicit algebraic argument that the m -dependent part $f(m, N) = m(N-m)/2$ is *metric-independent*—a consequence of $\mathbb{Z}_N$ representation theory alone—while only the additive constant C_1 carries the geometric information. A specialist who worked through the Boatto–Cabral calculation would likely have noticed this; the value here is in making the separation explicit and, more substantively, in extending the framework to the hyperbolic plane (§4.3), where $N_{\text{crit}} > 7$ and systematic stability tables do not appear to be in the existing literature.

4.1 The Riemannian Havelock identity

Boatto and Cabral [2003] computed numerically that the constrained Lagrangian eigenvalues on the sphere satisfy

$$\lambda_m = C_1(\varphi_0) - \frac{m(N-m)}{2}, \quad (15)$$

with C_1 depending on the colatitude φ_0 but $f(m, N) = m(N-m)/2$ unchanged from the planar value. The following proof makes explicit *why* $f(m, N)$ is invariant: both ingredients of its derivation (the anti-symmetry of the sine sum forcing $H_{rt}^{(m)} = 0$, and the $\mathbb{Z}_N$ orthogonality of characters) are properties of the discrete rotation group, not of the specific Green's function. Numerically: $C_2 = 1.000$ at all tested colatitudes to four decimal places; eigenvalue pair sums $\lambda_m^+ + \lambda_m^- = 2C_1$ are constant across all modes at each colatitude.

The universality follows from the proof structure of Appendix A: the Havelock identity $T_m = m(N-m)/2$ is derived from (i) the algebraic factorization $|\sum_{k=0}^{m-1} \omega^{pk}|^2$ and (ii) the orthogonality of $\mathbb{Z}_N$ characters, both of which are properties of the discrete rotation group $\mathbb{Z}_N$ —not of the specific Green's function. The Green's function enters only through: $H_{rt}^{(m)} = 0$ (anti-symmetry of the sine sum, requiring only that G depends on geodesic distance) and $H_{tt}^{(m)} + H_{rr}^{(m)} = \delta$, where $\delta = -\sum_p \nabla_{\text{LB}}^2 h(d_p)/(N-1)$ is a surface-dependent constant *independent of the Fourier mode m* .

Derivation of the trace formula. Let $\mathbf{M}_p$ be the 2×2 Hessian matrix of $h(d_{0p})$ with respect to the position of vortex 0, in the local (radial, tangential) frame; its trace is $\text{tr}[\mathbf{M}_p] = \nabla_{\text{LB}}^2 h(d_p)$. The standard $\mathbb{Z}_N$ -Fourier block decomposition (see Havelock 1931, §2) gives, for each component $\alpha, \beta \in \{r, t\}$:

$$H_{\alpha\beta}^{(m)} = \underbrace{-\sum_{p=1}^{N-1} M_{p,\alpha\beta}}_{\text{self-interaction at 0}} + \sum_{p=1}^{N-1} M_{p,\alpha\beta} \cos(2\pi pm/N) = \sum_{p=1}^{N-1} M_{p,\alpha\beta} (\cos(2\pi pm/N) - 1).$$

The self-interaction term $-\sum_p M_{p,\alpha\beta}$ is the negative sum of all off-diagonal blocks, forced by the translational zero mode of H (total gradient vanishes at the equilibrium). Taking $\alpha = r, \beta = r$ plus $\alpha = t, \beta = t$:

$$H_{rr}^{(m)} + H_{tt}^{(m)} = \sum_{p=1}^{N-1} \nabla_{\text{LB}}^2 h(d_p) (\cos(2\pi pm/N) - 1). \quad (16)$$

The coefficient $(\cos(2\pi pm/N) - 1)$ has the same “ $c_m = -1$ ” structure for every mode and every pair: it arises entirely from the self-interaction bookkeeping, not from any property of the Green's function.

Mode-independence. Equation (16) gives $H_{rr}^{(m)} + H_{tt}^{(m)} = [\sum_p \nabla_{\text{LB}}^2 h(d_p) \cos(2\pi pm/N)] - [\sum_p \nabla_{\text{LB}}^2 h(d_p)]$. On any compact 2D Riemannian manifold, the Green's function G of the Laplace–Beltrami operator satisfies $\nabla_{\text{LB},x}^2 G(x, y) = 1/\text{Vol}(M)$ for $x \neq y$ (constant, independent of x, y , or separation). Hence $\nabla_{\text{LB}}^2 h(d_p) = c$ is the *same constant* for every pair p , and $\sum_p \nabla_{\text{LB}}^2 h(d_p) \cos(2\pi pm/N) = c \sum_{p=1}^{N-1} \cos(2\pi pm/N) = -c$, which is independent of m . Therefore $H_{rr}^{(m)} + H_{tt}^{(m)} = -c - (N-1)c = -Nc \equiv \delta$ is mode-independent. On the plane ($\text{Vol} = \infty$), $c = 0$, so $\delta = 0$ (exact tracelessness); on curved

surfaces $c = 1/\text{Vol}(M) \neq 0$, giving $\delta = -N/\text{Vol}(M)$, which shifts C_1 without affecting $f(m, N)$. On the sphere of radius R : $\text{Vol} = 4\pi R^2$, so $\delta = -N/(4\pi R^2)$, consistent with the colatitude-dependent C_1 of Section 4.2.

The stability threshold N_{crit} is therefore determined by $C_1(\text{surface}) \geq \max_m m(N-m)/2$: a condition involving the *universal* function f and a single surface-dependent constant. On the plane: $C_1 = N-1$, giving $N_{\text{crit}} = 7$ (the largest integer with $(N-1) \geq \lfloor N/2 \rfloor (N - \lfloor N/2 \rfloor)/2$). On the sphere: $C_1(\varphi_0) < N-1$ for $\varphi_0 > 0$, giving the colatitude-dependent threshold of Section 4.2.

The value $C_1 = N-1$ on the plane has a geometric origin: for $h' = -1/d$ (the logarithmic force law), each vortex pair $(0, k)$ contributes the *same* amount $-1/(2R)$ to the gradient $\partial H/\partial x_0$, regardless of the separation k . This is because $h'(d_k)(x_0 - x_k)/d_k = -(1 - \cos(2\pi k/N))/(4R \sin^2(\pi k/N)) = -1/(2R)$: the geometric factor $(1 - \cos)$ cancels against the denominator $\sin^2 = (1 - \cos)/2$. The cancellation is the *scale invariance* of $h' = -1/d$: the force law $h' \propto 1/d$ precisely compensates the d -dependence of the chord geometry, making each pair's contribution independent of the separation. This property characterizes the logarithm *uniquely*: among radially symmetric pairwise interactions on $\mathbb{R}^2$, $h = -\ln|z-w|$ is the unique function satisfying $h(d\lambda) = h(d) + h(\lambda)$ (additivity under rescaling—Cauchy's functional equation with continuity). This additivity under the dilation subgroup $z \mapsto \lambda z$ (for which $|f(z) - f(w)| = |\lambda| |z-w|$) is Cauchy's multiplicative functional equation $h(ab) = h(a) + h(b)$, whose unique continuous solution is $h = c \ln$.

4.2 Curved 2D surfaces: the spherical stability boundary

On the 2-sphere, the point vortex interaction $h(\theta) = -\ln(2 \sin(\theta/2))$ satisfies $h' < 0$ and $(\sin \theta \cdot h')' = \frac{1}{2} \sin \theta > 0$ (class (iii)). While $H''(0) > 0$ along the spiral at all tested colatitudes (the polygon is a spiral-direction energy minimum even without the Lagrange shift), the *constrained* Hessian tells a different story.

The angular impulse on the sphere is $L = \sum \cos \theta_k$, with Hessian $\partial^2 L / \partial \theta_k^2 = -\cos \varphi_0$ at colatitude φ_0 . The Lagrangian shift from μ_L is therefore proportional to $\cos \varphi_0$: full near the pole ($\varphi_0 \rightarrow 0$), vanishing at the equator ($\varphi_0 = 90^\circ$). This latitude-dependent shift produces a *colatitude-dependent stability boundary*:

N	3	4	5	6	7	≥ 8
φ_{crit}	60.0°	48.2°	41.4°	25.8°	0°	0°

The N -gon is a constrained energy minimum on the sphere if and only if $\varphi_0 < \varphi_{\text{crit}}(N)$. The dimensionless eigenvalue $\bar{\lambda} = \sin^2 \varphi_0 \cdot \lambda$ recovers the Havelock values to within 1% at $\varphi_0 = 3^\circ$ (N=5: $\bar{\lambda} = 0.99$, Havelock = 1.0; N=6: 0.49 vs 0.5; N=8: -1.01 vs -1.0).

Saturn's hexagon at 76° N ($\varphi_0 = 14^\circ$) lies inside $\varphi_{\text{crit}}(6) = 26.6^\circ$: the hexagonal ring is in the stable zone. If the hexagon were at 60° N ($\varphi_0 = 30^\circ > 26.6^\circ$), it would cross the boundary and the pattern would not persist. Jupiter's octagon ($N = 8$) is unstable at all colatitudes, consistent with the need for a central vortex.

Derivation of $C_1(\mathbb{S}^2)$ and exact thresholds. For N vortices on a latitude circle at colatitude φ_0 , the great-circle separation satisfies $2 \sin(\gamma_{jk}/2) = 2 \sin \varphi_0 |\sin(\Delta \phi_{jk}/2)|$, so the spherical pair interaction decomposes as

$$-\ln(2 \sin(\gamma_{jk}/2)) = -\ln(2 |\sin(\Delta \phi_{jk}/2)|) - \ln(\sin \varphi_0).$$

The first term is the flat Thomson energy in the azimuthal variable ϕ ; its Hessian along azimuthal Fourier modes gives the Havelock eigenvalues $(N-1)/2 - m(N-m)/2$ exactly as in the flat case. The second term $-\binom{N}{2} \ln(\sin \varphi_0)$ is purely radial and enters through the Lagrange multiplier.

The angular impulse on $\mathbf{S}^2$ is $L = \sum \cos \varphi_k$. At the ring, $\partial^2 L / \partial \varphi_k^2 = -\cos \varphi_0$, so the Lagrange shift $\mu_L \cdot \partial^2 L / \partial \varphi_k^2$ is proportional to $\cos \varphi_0$: maximal near the pole, vanishing at the equator. Computing μ_L from the stationarity condition and combining with the radial Hessian of the pair interaction yields the curvature coefficient (cf. Boatto and Cabral 2003):

$$C_1(\mathbf{S}^2, \xi) = \frac{(N-1)(1-\xi)}{1+\xi} = (N-1) \cos \varphi_0, \quad (17)$$

where $\xi = \tan^2(\varphi_0/2)$ is the stereographic parameter and $\cos \varphi_0 = (1-\xi)/(1+\xi)$. Marginal stability $C_1 = m(N-m)/2$ therefore reduces to

$$\cos \varphi_{\text{crit}} = \frac{m(N-m)}{2(N-1)}, \quad m = \lfloor N/2 \rfloor, \quad (18)$$

which is *always rational*. The threshold in ξ is

$$\xi_{\text{crit}}(N) = \frac{(N-1) - m(N-m)/2}{(N-1) + m(N-m)/2} \in \mathbb{Q}. \quad (19)$$

Explicit values:

N	$m(N-m)/2$	$\cos \varphi_{\text{crit}}$	ξ_{crit}	φ_{crit}	stable iff
3	1	1/2	1/3	60.0°	$\varphi_0 < 60^\circ$
4	2	2/3	1/5	48.2°	$\varphi_0 < 48.2^\circ$
5	3	3/4	1/7	41.4°	$\varphi_0 < 41.4^\circ$
6	9/2	9/10	1/19	25.8°	$\varphi_0 < 25.8^\circ$
≥ 7	$\geq m(N-m)/2 > N-1$	< 0	—	—	never

The thresholds $\xi_{\text{crit}} \in \{1/3, 1/5, 1/7, 1/19\}$ are exact rational numbers even though the Green's function $h(\theta) = -\ln(2 \sin(\theta/2))$ is transcendental—the rationality is a consequence of the linear dependence of C_1 on $\cos \varphi_0$ ((17)) combined with the rational marginal condition (18).

Remark 4 (Pattern $\xi_{\text{crit}} = 1/(2N-3)$ and its break at $N = 6$). *For $N = 3, 4, 5$ one has $m(N-m)/2 = N-2$ exactly (since $m = \lfloor N/2 \rfloor = (N-1)/2$ or $N/2$ gives the same result), so the numerator $(N-1) - (N-2) = 1$ and denominator $(N-1) + (N-2) = 2N-3$, yielding $\xi_{\text{crit}} = 1/(2N-3)$. At $N = 6$ this pattern breaks: the critical mode has $m(N-m)/2 = 9/2 \neq N-2 = 4$, because the hexagonal mode carries a larger Havelock eigenvalue ($\lambda_{\min} = 1/2$ vs. 1 for $N \leq 5$). Consequently $\xi_{\text{crit}}(6) = 1/19$, not the $1/(2 \cdot 6 - 3) = 1/9$ that the pattern would predict.*

The notation $\bar{\lambda} = \sin^2 \varphi_0 \cdot \lambda$ (line 938–941) converts the dimensionful eigenvalue λ to the dimensionless form used in the numerical table above; the exact ξ -threshold (19) is independent of this rescaling.

The colatitudes in the summary table above are the exact values $\varphi_{\text{crit}} = \arccos(m(N-m)/(2(N-1)))$; for $N = 6$, $\cos \varphi_{\text{crit}} = 9/10$.

Code: `algebraic_thresholds.sphere_stability_threshold(N)` and `sphere_threshold_table()` implement these exact computations using Python's `Fraction` arithmetic; `test_algebraic_thresholds` verifies the C_1 identity (18) by exact rational arithmetic for $N = 3, \dots, 6$.

On the flat torus, $H''(0) < 0$ approaches the planar value for polygons small relative to the domain, with periodic corrections entering at the domain scale.

4.3 The hyperbolic plane and Gaussian curvature

Kimura [1999] and others have studied point vortex dynamics on surfaces of constant curvature; the Hamiltonian for the hyperbolic plane is known. Systematic stability tables for N -gons on $\mathbf{H}^2$ do not appear to be in the literature, and the results of this subsection are likely the most novel part of Section 4.

The threshold N_{crit} on any surface is the largest N with $C_1(\text{surface}) \geq \max_m m(N-m)/2$. Gaussian curvature K is the control parameter:

Curvature	N_{crit}	Mechanism
$K > 0$ (sphere)	< 7	Positive curvature destabilizes
$K = 0$ (plane)	$= 7$	Critical transition
$K < 0$ (hyperbolic)	> 7	Negative curvature stabilizes

Remark 5 (Order of the curvature correction). *The curvature enters N_{crit} at $O(K)$, not $O(K^2)$: it acts through the surface Green's function via the Havelock coefficient $C_1(\text{surface}, \xi)$ (equations (17), (20)), which is linear in K at leading order. On a compact surface the Gauss–Bonnet constraint $\int K \, dA = 2\pi\chi$ enters only through C_1 ; no additional $O(K^2)$ self-energy correction arises for point vortices (such a correction would require finite-area patches, where it is $O(\varepsilon^2)$ and topology-independent).*

On the hyperbolic plane with curvature radius a , numerical computation (Poincaré disk model, exact Green's function $h = -\ln \sinh(d_{\mathbf{H}^2}/(2a))$, corrected Lagrange multiplier for the $\text{SO}(2)$ moment map $J = \sum_k (a^2 + r_k^2)/(a^2 - r_k^2)$) gives:

R/a	N_{crit}	$7 \rightarrow N$ transition
0.30–0.50	7	(no change)
≈ 0.52	8	$7 \rightarrow 8$ transition
0.65–0.75	9	
0.80	10	
0.90	11	
1.0	12	
1.2	16	
≥ 1.5	≥ 24	(tested up to $N = 24$)

The Riemannian Havelock identity $\lambda_m = C_1(R/a) - m(N-m)/2$ is verified to hold (computed C_1 values are mode-independent to numerical precision $\sim 10^{-5}$). The exact $\mathbf{H}^2$ Green's function yields the closed-form expression (derived in Appendix B)

$$C_1(\mathbf{H}^2, \xi) = \frac{(N-1)(1+\xi^2)}{(1-\xi)^2}, \quad \xi = \frac{r_E^2}{a^2} = |K| r_E^2, \quad (20)$$

where r_E is the Euclidean ring radius in the Poincaré disk and $K = -1/a^2$. Since $(1 + \xi^2)/(1 - \xi)^2 > 1$ for all $\xi > 0$, one has $C_1(\mathbf{H}^2) > N-1$ for all $R/a > 0$, confirming the stabilizing effect of negative curvature.

N_{crit} increases through integer jumps as R/a grows, with the $7 \rightarrow 8$ transition at $R/a \approx 0.52$ (more precisely, the $m = \lfloor N/2 \rfloor$ eigenvalue of the $N = 8$ ring crosses zero near this value). We *conjecture* that $N_{\text{crit}}(R/a)$ is non-decreasing (verified for $R/a \leq 2$; a plausible mechanism is the transition from class (ii) at $R \ll a$ to class (i) at $R \gg a$, but this does not constitute a proof because the Lagrange multiplier also depends on R/a in a non-trivial way).

Near- $K = 0$ behavior. Expanding (20) for $N = 7$, $m = 3$ (see Appendix B for the algebraic simplification), the $m = 3$ eigenvalue satisfies

$$\lambda_3(N = 7, \xi) = \frac{12\xi}{r_E^2(1 - \xi)^2} = \frac{12|K|}{(1 - \xi)^2} = 12|K| + O(K^2 r_E^2) > 0. \quad (21)$$

Negative curvature stabilizes the $N = 7$ ring at *first* order in $|K|$: $N_{\text{crit}}(K)$ stays at 7 for all $K \leq 0$ in the sense that the formerly-neutral mode becomes strictly stable.

The $7 \rightarrow 8$ transition is governed by the $N = 8$ ring reaching marginal stability. Setting $C_1(\mathbf{H}^2, \xi) = m(N-m)/2|_{m=4} = 8$ gives $7(1 + \xi^2)/(1 - \xi)^2 = 8$, which simplifies to

$$\xi^2 - 16\xi + 1 = 0 \implies \xi^* = 8 - 3\sqrt{7} \approx 0.0627, \quad (22)$$

exact. In terms of geodesic ring radius ρ , this corresponds to $\rho/a \approx 0.52$ (or $|K|\rho^2 \approx 0.27$). The coefficient

$$\gamma \equiv \frac{1}{(\xi^*)^2} = 127 + 48\sqrt{7} \approx 254 \quad (23)$$

carries an algebraic interpretation. The ring of integers of $\mathbb{Q}(\sqrt{7})$ has fundamental unit $\varepsilon = 8 + 3\sqrt{7}$, since $(8 + 3\sqrt{7})(8 - 3\sqrt{7}) = 64 - 63 = 1$. Therefore $\xi^* = 8 - 3\sqrt{7} = \varepsilon^{-1}$ is the *inverse of the fundamental unit*—the unique unit below 1 in $\mathbb{Z}[\sqrt{7}]$ —and $\gamma = (\xi^*)^{-2} = \varepsilon^2$. The verification identity $(8 - 3\sqrt{7})^2(127 + 48\sqrt{7}) = 1$ is not a computational coincidence: it is the norm-1 property $\varepsilon^{-2} \cdot \varepsilon^2 = 1$ in $\mathbb{Z}[\sqrt{7}]$.

The field $\mathbb{Q}(\sqrt{7})$ arises because $N_{\text{crit}} = 7$ enters the marginal condition; this is a mathematical consequence, not an independent observation. The substantive content is the *palindromic structure*: the threshold polynomial has the form $A\xi^2 + B\xi + A = 0$ (same leading and trailing coefficient), which forces $\xi_1 \xi_2 = A/A = 1$ —the two roots are multiplicative inverses, so ξ^* is automatically an algebraic *unit* (an element of norm 1 in $\mathbb{Z}[\sqrt{7}]$). This palindromic property is not generic: it reflects the inversion symmetry $\xi \leftrightarrow 1/\xi$ of the Poincaré disk model, which in turn is the restriction to the real axis of the conformal automorphism $z \mapsto a^2/\bar{z}$ of the disk. The stability threshold is a unit *because* the hyperbolic energy is conformally covariant (Proposition 2): the palindromic polynomial encodes the same Möbius symmetry at the level of algebraic number theory.

General algebraic field pattern ($N \geq 8$). The $7 \rightarrow 8$ transition is not an isolated result: the $N \rightarrow (N+1)$ transition threshold on $\mathbf{H}^2$ is always the smaller positive root of the palindromic quadratic

$$A\xi^2 + B\xi + A = 0, \quad A = (N-1) - \frac{m(N-m)}{2}, \quad B = m(N-m), \quad m = \lfloor N/2 \rfloor. \quad (24)$$

Since $A < 0$ for $N \geq 8$ and $B > 0$, the smaller positive root is $\xi^* = (B/2 - \sqrt{(B/2)^2 - A^2}) / |A|$, and its field extension is determined by the squarefree part of the discriminant $(B/2)^2 - A^2 \in \mathbb{Z}$:

N	ξ^* (approx.)	Field	D	ξ^* (exact form)
7	0	$\mathbb{Q}$	1	0 (marginal)
8	0.0627	$\mathbb{Q}(\sqrt{7})$	7	$8 - 3\sqrt{7}$
9	0.1010	$\mathbb{Q}(\sqrt{6})$	6	$(5 - \sqrt{6})/\sqrt{6} \dots$
10	0.1429	$\mathbb{Q}$	1	$1/7$
11	0.1716	$\mathbb{Q}(\sqrt{2})$	2	
12	0.1952	$\mathbb{Q}(\sqrt{11})$	11	
13	0.2162	$\mathbb{Q}(\sqrt{10})$	10	
14	0.2355	$\mathbb{Q}(\sqrt{13})$	13	
15	0.2533	$\mathbb{Q}(\sqrt{3})$	3	
16	0.2699	$\mathbb{Q}(\sqrt{15})$	15	

The discriminant's squarefree part D follows the pattern

$$D = \begin{cases} \text{sq_free}(N-1) & N \text{ even,} \\ \text{sq_free}(N-3) & N \text{ odd,} \end{cases} \quad (25)$$

for all $N \geq 8$. This is not merely a computational observation: it follows from the factorization of the discriminant $\Delta = B^2 - 4A^2 = 4(N-1)[m(N-m) - (N-1)]$. For even N ($m = N/2$): $m(N-m) = N^2/4$, so $\Delta/4 = (N-1)(N-2)^2/4$, and $\text{sq_free}(\Delta/4) = \text{sq_free}(N-1)$ (since $(N-2)^2$ and 4 are perfect squares). For odd N ($m = (N-1)/2$): $m(N-m) = (N^2-1)/4$, so $\Delta/4 = (N-1)^2(N-3)/4$, and $\text{sq_free}(\Delta/4) = \text{sq_free}(N-3)$.

The $N = 10$ case is the unique rational threshold: $N-1 = 9 = 3^2$ is a perfect square, so $D = 1$ and $\xi^*(10) = 1/7 \in \mathbb{Q}$, the same squarefree-discriminant mechanism that makes $\xi^*(8) = 8 - 3\sqrt{7} \in \mathbb{Q}(\sqrt{7})$ an inverse fundamental unit. Every even transition lives in at most a degree-2 extension of $\mathbb{Q}$; the tower terminates at $\mathbb{Q}$ itself whenever $N-1$ is a perfect square.

Code: `algebraic_thresholds.h2_stability_threshold(N)`, `h2_threshold_polynomial(N)`, and `h2_threshold_table()` implement these computations; `test_algebraic_thresholds.py` verifies the field-pattern table for $N = 8, \dots, 16$ and the C_1 -condition $C_1(\mathbf{H}^2, \xi^*) = m(N-m)/2$ for each N to 10^{-8} relative error.

The coefficient γ gives the curvature scale at which the polygon threshold is reached: the $7 \rightarrow 8$ jump occurs at $|K| r_E^2 = \xi^* = \gamma^{-1/2}$. There is no perturbative formula $N_{\text{crit}} \approx 7 + c|K| r_E^2$; the threshold (22) is a *finite*, non-perturbative condition, and N_{crit} is an integer that jumps discontinuously.

For any fixed ratio R/a of ring radius to curvature radius, the Gaussian curvature K controls the threshold: $K > 0$ reduces N_{crit} below 7, $K < 0$ raises it above 7, and $K = 0$ gives exactly 7. The number 7 is the stability threshold at zero Gaussian curvature *at the scale of the polygon ring*. $K = 0$ is the *least stable point*: the $N = 7$ ring is marginal at $K = 0$ and strictly stable for any $K < 0$. The marginality is an artifact of $K = 0$ exactly; $N_{\text{crit}} = 7$ is the value for the dilation-invariant logarithmic interaction (Remark 2), and any nonzero negative curvature eliminates the marginal mode at first order in $|K|$ via (21).

Physical realization. The hyperbolic plane is no longer a purely mathematical object. Kollár et al. [2019] demonstrated that microwave resonators arranged in a tiling of the Poincaré disk implement the hyperbolic lattice Hamiltonian in circuit quantum electrodynamics, with negative effective curvature controlled by the lattice geometry. Lenggenhager et al. [2022] subsequently implemented hyperbolic space on a classical circuit board, showing that band structure and topological properties native to $\mathbf{H}^2$ are experimentally accessible. These platforms realize, in a controlled laboratory setting, the geometry that Section 4.3 analyzes.

The stability prediction translates directly into an experimental observable: *a ring of N vortex-like excitations on a hyperbolic lattice with ring-to-curvature ratio $R/a \approx 1$ should be linearly stable up to $N = 12$, without requiring any central defect.* In flat geometry, a 12-vortex ring is linearly unstable at every curvature-free configuration and can only be stabilized by a sufficiently strong central vortex (the Thomson–Havelock criterion requires a central circulation $\kappa_0/\kappa \geq |\kappa_{\text{crit}}(12)|$, which is large). The hyperbolic curvature replaces the stabilizing role of the central vortex: the geometry itself provides the restoring force. A predicted $N = 12$ stable ring with no center is impossible in flat geometry and would be a direct signature of hyperbolic curvature acting on vortex dynamics.

What would need to be realized is the hyperbolic-plane Green’s function for vortex interactions, $h(r_{\mathbf{H}^2})$, where $r_{\mathbf{H}^2}$ is the geodesic distance [Kimura, 1999]. In a 2D superfluid hosted on a hyperbolic lattice—for example, a Bose–Hubbard model implemented in circuit QED—the vortex interaction would inherit the hyperbolic geometry. Equivalently, topological defects in a photonic condensate on a Poincaré-disk resonator array would interact via the hyperbolic Hamiltonian. Neither realization exists yet. The prediction $N_{\text{crit}} = 12$ at $R/a = 1$ (and $N_{\text{crit}} \geq 14$ at $R/a = 2$) is a falsifiable quantitative target for such an experiment.

Three conditions are required for the hyperbolic Hamiltonian $h = -\ln \sinh(d_{\mathbf{H}^2}/2a)$ to emerge as the effective vortex interaction. First, the platform must support a 2D superfluid order parameter $e^{i\theta}$ on the hyperbolic lattice, so that vortices appear as topological defects with winding number ± 1 in θ . Second, in the long-wavelength limit the phase field must satisfy the hyperbolic Laplace–Beltrami equation $\Delta_g \theta = 0$, whose Green’s function on $\mathbf{H}^2$ is precisely h [Kimura, 1999]; this follows automatically once the lattice dispersion relation is hyperbolic. Third, the curvature radius a and ring radius R must both be independently controllable: a is set by the lattice geometry (fixed at fabrication, as demonstrated by Kollár et al. 2019), while R can be tuned by pinning vortex excitations to engineered impurities or to the boundary of the resonator array. In a Bose–Hubbard model on a Poincaré-disk resonator array these three conditions are simultaneously achievable: the superfluid phase satisfies the first, the hyperbolic dispersion relation the second, and current fabrication control over impurity placement is sufficient for the third. A realized system of this type would allow direct measurement of the $N_{\text{crit}}(R/a)$ curve—a signature inaccessible in flat geometry—by counting the largest stable vortex ring as R/a is varied.

5 The three-scale physical mechanism

5.1 The three-scale mechanism

The physical picture is:

Scale	Question	Mechanism
Macro	Why do N coherent vortices form?	Onsager + inverse cascade
Meso	Which N is selected?	Onsager max- N (eq. (26)) + Thomson upper bound + Rossby n^* (jets only)
Micro	Why regular polygon?	Arnold/Thomson stability

The Onsager argument [Onsager, 1949] operates at the macro scale: the inverse cascade and negative-temperature statistics explain why same-sign vorticity concentrates into N coherent clusters. *This is a physical hypothesis for planetary atmospheres, not a derived result, and is the key assumption that lifts $N = 8$ over $N = 7$ in the Jupiter selection (Proposition 9).* The Siegelman et al. [2022] simulations—2D turbulence on idealized polar caps with stochastic forcing—are *consistent with* the Onsager hypothesis: they develop an inverse cascade, produce $N = 8$ (north) and $N = 5$ (south) vortex crystals, and reach a maximum- H state within the Thomson-stable set. However, these simulations use 2D barotropic dynamics with prescribed forcing statistics, not Jupiter’s actual three-dimensional, moist-convective, differentially rotating environment. The agreement is therefore evidence *consistent with* the Onsager picture, not evidence *for* it in the strict sense.

Whether Jupiter’s polar cyclones are actually in or near an Onsager negative-temperature equilibrium—rather than being dynamically maintained metastable states sustained by ongoing moist convection—is an open empirical question. A direct test would require measuring the energy spectrum at the polar scale, which is not yet available from Juno.

The polygon arrangement is then determined by the meso-scale energy landscape: the regular N -gon minimizes the interaction energy at constant angular momentum (Theorem 5). This is a deterministic stability result, not a statistical one.

5.2 Complementary wavenumber selection

Proposition 8 (Complementary selection). *The observed polygon wavenumber N is constrained by two complementary mechanisms, both arising from the logarithmic Green’s function:*

1. **Rossby stationarity** (for jet meanders): *the QGPV eigenvalue problem uniquely selects the wavenumber n^* of the stationary mode ($\omega = 0$). For Saturn: leading-order $n^* \approx 5.4$, full Rayleigh–Kuo eigenvalue $n^* = 6$.*
2. **Thomson stability** (for discrete cyclone rings): *$\kappa_0/\kappa \geq \kappa_{\text{crit}}(N)$ is a necessary condition—an upper bound $N \leq N_{\text{max}}(\kappa_0/\kappa)$, not a unique selection. The energy monotonicity $H_{N+1} > H_N$ (Proposition 9) then identifies $N = N_{\text{max}}$ as the unique maximum-energy stable configuration; any dynamics that preferentially populates higher-energy states selects this N .*

For Saturn, n^* is determined by the Rayleigh–Kuo condition for stationary waves on a jet. All quantities are evaluated in the frame rotating with Saturn’s System III period—the frame co-rotating with the deep interior. In this frame the background wind at the hexagonal jet is $U_{\text{max}} \approx 120$ m/s eastward. At the center of a Gaussian jet, the effective PV gradient is $\beta_{\text{eff}} = \beta + U_{\text{max}}/\sigma_{\text{jet}}^2$, where the jet curvature term dominates bare β by a factor ~ 13 for Saturn [Pedlosky, 1987, Vallis, 2017]. The leading-order stationary

wavenumber is $n^* = R \cos \varphi \sqrt{\beta_{\text{eff}}/U_{\text{max}}} \approx 5.4$. This is a leading-order estimate; the full numerical Rayleigh–Kuo eigenvalue computation yields $n^* = 6$ [Sánchez-Lavega et al., 2014], with the difference reflecting finite jet-width corrections (see Section 5.4.3 for the detailed derivation). Thomson allows $N \leq 7$; both criteria are satisfied at $N = 6$.

For Jupiter’s north pole ($N = 8$): the cyclones are discrete vortex clusters, not a jet meander; Rossby stationarity gives $n^* \approx 0.3$ (not physically relevant). *Caveat*: the 2D point vortex model requires the cyclone roots to extend through a barotropic column much deeper than the ring radius. Juno gravity data constrains Jupiter’s differential rotation to $\sim 3,000$ km depth [Kaspi et al., 2018], roughly 4% of R_J —making Jupiter the least observationally supported case for the 2D assumption among our applications. The model is nevertheless applicable if the polar cyclones are confined to a deep weather layer ($H_{\text{eff}} \sim 3,000$ km, $R_d \gg R$) or to a shallow tropospheric layer with reduced gravity; both scenarios are consistent with the Juno data (see Section 5.5 for quantitative limits).

Thomson stability provides an *upper bound*: $\kappa_{\text{crit}}(9) = 1 > \kappa_0/\kappa$, so $N = 9$ is unstable; $\kappa_{\text{crit}}(8) = \frac{1}{2} \leq \kappa_0/\kappa$, so $N = 8$ is stable [Adriani et al., 2018]. But $N = 7$ is also stable ($\kappa_{\text{crit}}(7) = 0$), so stability alone does not select $N = 8$ over $N = 7$. The selection requires one additional ingredient: a reason for the system to prefer higher-energy configurations among the stable set.

Proposition 9 (Energy monotonicity and maximum- N selection). *Let $\Gamma = N\kappa$ be the total ring circulation, $L = N\kappa R^2$ the angular impulse (so R is fixed when Γ and L are fixed), and $r = \kappa_0/\kappa$ held fixed as N varies (both the ring and central circulations scale proportionally). Denote by r_{core} the vortex core radius—the minimum separation at which the point-vortex model is valid; equivalently, r_{core} is the unit of length in which the logarithm is taken, so that $\ln R > 0$ iff $R > r_{\text{core}}$. Then*

$$H_{N+1} - H_N = \Gamma^2 \left[\frac{2r - 1}{2N(N+1)} \ln R + \frac{1}{2} \left(\frac{\ln N}{N} - \frac{\ln(N+1)}{N+1} \right) \right]. \quad (26)$$

For $r \geq \frac{1}{2}$ and $R > r_{\text{core}}$, both terms are non-negative (first: $2r - 1 \geq 0$ and $\ln R > 0$; second: $\ln N/N$ is strictly decreasing for $N > e \approx 2.7$), so $H_{N+1} > H_N$ for all $N \geq 3$. Therefore the N -gon ring energy is strictly monotone in N : the maximum- H configuration within the Thomson-stable set is the ring with the largest N :

$$N_{\text{max}}(\kappa_0/\kappa) \equiv \max \{ N' : \kappa_{\text{crit}}(N') \leq \kappa_0/\kappa \}.$$

Any dynamics that preferentially populates higher-energy stable configurations—whether Onsager negative-temperature equilibrium ($\beta < 0$, where the Boltzmann weight $e^{-\beta H}$ increases with H), sequential vortex injection (adding cyclones until Thomson instability prevents $N+1$), or merger of smaller vortex clusters—selects $N = N_{\text{max}}$. The mathematical content of the proposition is the monotonicity $H_{N+1} > H_N$; the physical mechanism by which the system reaches N_{max} is a separate question (see the discussion below and Section 5).

The stability threshold $\kappa_{\text{crit}}(8) = \frac{1}{2}$ is exactly the value at which $H_{N+1} > H_N$ becomes unconditional (i.e. R -independent): at $r = \frac{1}{2}$, term 1 vanishes identically and term 2 remains strictly positive.

Proof. Use the Hamiltonian $H = -\kappa^2 \sum_{j < k} \ln |z_j - z_k| - \kappa_0 \kappa \sum_j \ln |z_j|$ with $\kappa = \Gamma/N$ and $\kappa_0 = r\kappa = r\Gamma/N$.

Step 1: Product formula. For the regular N -gon $z_j = Re^{2\pi ij/N}$, the chord length between vortices j and k is $|z_j - z_k| = 2R \sin(\pi|j - k|/N)$. By $\mathbb{Z}_N$ symmetry the pair sum equals $N/2$ times the sum from one vertex:

$$\sum_{j < k} \ln |z_j - z_k| = \frac{N}{2} \sum_{s=1}^{N-1} \ln(2R \sin(\pi s/N)) = \frac{N}{2} [\ln N + (N-1) \ln R].$$

Here we used $\sum_{s=1}^{N-1} \ln(2 \sin(\pi s/N)) = \ln N$, which follows from evaluating the polynomial identity $(z^N - 1)/(z - 1) = \prod_{k=1}^{N-1} (z - e^{2\pi ik/N})$ at $z = 1$: the product of moduli is $\prod_{s=1}^{N-1} |1 - e^{2\pi is/N}| = \prod_{s=1}^{N-1} 2 \sin(\pi s/N) = N$.

Step 2: Central vortex term. Since $|z_j| = R$ for all j : $\sum_j \ln |z_j| = N \ln R$.

Step 3: Closed form for H_N . Substituting Steps 1–2 with $\kappa = \Gamma/N$, $\kappa_0 = r\Gamma/N$:

$$H_N = -\frac{\Gamma^2}{2N} \ln N - \Gamma^2 \left(\frac{N-1}{2N} + \frac{r}{N} \right) \ln R. \quad (27)$$

Step 4: Difference $H_{N+1} - H_N$. Writing H_{N+1} from (27) with $N \rightarrow N+1$ and subtracting:

$$H_{N+1} - H_N = -\frac{\Gamma^2}{2} \left[\frac{\ln(N+1)}{N+1} - \frac{\ln N}{N} \right] - \Gamma^2 \ln R \left[\frac{N/2 + r}{N+1} - \frac{(N-1)/2 + r}{N} \right].$$

The coefficient of $\ln R$ simplifies by partial fractions:

$$\frac{N/2 + r}{N+1} - \frac{(N-1)/2 + r}{N} = \frac{N^2 - (N-1)(N+1)}{2N(N+1)} - \frac{r}{N(N+1)} = \frac{1 - 2r}{2N(N+1)},$$

where we used $N^2 - (N-1)(N+1) = 1$. Negating and rearranging gives (26).

Step 5: Sign of each term. *Term 2:* $f(x) = \ln x/x$ satisfies $f'(x) = (1 - \ln x)/x^2 < 0$ for $x > e$, so $\ln N/N > \ln(N+1)/(N+1)$ for all $N \geq 3$.

Term 1: $2r - 1 \geq 0$ for $r \geq \frac{1}{2}$; and $\ln R > 0$ for $R > r_{\text{core}}$. Hence both terms are non-negative, and term 2 is strictly positive, giving $H_{N+1} > H_N$. $\square$

Remark 6 (Physical status of the fixed- r assumption). *The assumption $r = \kappa_0/\kappa$ fixed as N varies means that as more cyclones appear in the ring (N increases), each individual ring vortex becomes weaker ($\kappa = \Gamma/N$) and the central vortex weakens proportionally ($\kappa_0 = r\Gamma/N$). This is non-trivial: the central polar cyclone is a distinct dynamical object from the ring cyclones and need not respond to changes in N .*

The assumption is justified when all vortices — ring and central — are drawn from the same turbulent condensation process, so their strengths are set by a common large-scale energy budget rather than by independent forcings. In the Onsager picture, the negative-temperature state is reached by the inverse cascade from the same vorticity reservoir, giving a fixed partition of total circulation between center and ring. If instead κ_0 is held fixed while $\kappa = \Gamma/N$ varies (independent central vortex), then $r = N\kappa_0/\Gamma$ grows with N and the monotonicity of $H_{N+1} > H_N$ strengthens further, so the maximum- N selection remains qualitatively intact. The marginal case is $\kappa_0 = 0$ (no central vortex): then $r = 0 < \frac{1}{2}$ and Term 1 in (26) is negative, so $H_{N+1} > H_N$ requires R large enough that Term 2 dominates. The proposition's $r \geq \frac{1}{2}$ condition is therefore specific to the regime in which the central vortex is sufficiently strong relative to the ring vortices.

The selection of $N = 8$ over $N = 7$ is sensitive to which assumption is made. Under fixed r : $H_{N+1} > H_N$ for all $N \geq 3$ (Proposition 9), so Onsager selects the Thomson-maximum $N_{\max}$. Under fixed κ_0 : r grows with N , which strengthens the monotonicity and gives the same $N_{\max}$. Under the intermediate assumption $\kappa_0 = \text{const}$ while Γ also varies: the result depends on how $\Gamma(N)$ scales, which is not observationally constrained. Thus the $N = 8$ selection is robust across the two limiting assumptions (fixed r and fixed κ_0) but relies on the Onsager framework providing the link between N , Γ , and κ_0 —a physical hypothesis, not a mathematical consequence.

Observationally, Jupiter’s central cyclone differs from the ring cyclones: it has a smaller diameter ($\sim 4,000$ km vs. $4,000$ – $7,000$ km; Adriani et al. 2018) and appears more axisymmetric. Whether it formed from the same inverse-cascade process as the ring cyclones, or was generated by a distinct polar convective mechanism, is not resolved by current Juno data.

For Jupiter north, $\kappa_0/\kappa \approx 1$ with roughly 30% observational uncertainty (Juno; Adriani et al. 2018), giving a conservative range $\kappa_0/\kappa \in [0.7, 1.3]$. Since $\kappa_{\text{crit}}(8) = \frac{1}{2}$ exactly, this entire range satisfies the stability condition for $N = 8$. However, $\kappa_{\text{crit}}(9) = 1$, which lies *inside* the uncertainty interval: if $\kappa_0/\kappa \geq 1$ (the upper half of the range), then $N = 9$ is also Thomson-stable and the Onsager argument would select $N_{\max} = 9$ instead. The conclusion $N_{\max} = 8$ therefore relies on $\kappa_0/\kappa < 1$, which is consistent with the central estimate but not guaranteed by the current observational precision. More precisely: $\kappa_0/\kappa \leq 1$ gives $N_{\max} = 8$ (observed); $\kappa_0/\kappa > 1$ gives $N_{\max} = 9$ (not observed). The observed $N = 8$ is thus evidence that $\kappa_0/\kappa \lesssim 1$, providing a theory-derived constraint on the central vortex strength that could be tested with improved Juno wind measurements.

For Jupiter’s south pole ($N = 5$): $\kappa_{\text{crit}}(N) = 0$ for all $N \leq 7$, so the Thomson bound alone allows up to $N = 7$ regardless of the central vortex. The observed $N = 5$ is not explained by Thomson stability. Gavriel and Kaspi [2021] argue that $N = 5$ follows from the β -drift / vortex-repulsion equilibrium at the south-pole colatitude, combined with the fact that south-pole cyclones are individually $\approx 2\times$ larger in linear size than north-pole cyclones [Adriani et al., 2018]: fewer large vortices fit at the equilibrium latitude. This is a packing constraint that lies outside the scope of the Thomson/Onsager framework developed here.

5.3 Eigenvalue branch disconnection (Saturn persistence)

The preceding subsection selected Saturn’s wavenumber $n = 6$ via Rossby stationarity. A natural follow-up: can smooth perturbations of the jet destroy the stationary wave, converting the hexagonal meander into an evanescent (decaying) mode? The answer is no, by a WKB branch-disconnection argument.

The Rayleigh–Kuo equation for stationary ($c = 0$) mode n on a zonal jet $U(y)$ reads $\hat{\psi}'' + V(\rho)\hat{\psi} = 0$, where $V = (\beta - U'')/U - n^2/R^2$ is the effective potential in log-polar coordinates. At the jet center ($y = 0$) of a Gaussian jet:

$$V(0) = \frac{\beta + U_{\max}/\sigma^2}{U_{\max}} - \frac{n^2}{R^2} = \frac{\beta_{\text{eff}}}{U_{\max}} - \frac{n^2}{R^2}.$$

For Saturn ($U_{\max} = 120$ m/s, $\sigma = 2.5 \times 10^6$ m, $\beta = 1.46 \times 10^{-12}$ m^{−1}s^{−1}, $n = 6$, $R = R_S \cos 76^\circ = 1.31 \times 10^7$ m): $V(0) = 1.72 \times 10^{-13} - 2.10 \times 10^{-13} = -3.8 \times 10^{-14}$ m^{−2} (negative; in dimensionless units $V \cdot R^2 \approx -6.5$, or $V \cdot \sigma^2 \approx -0.24$; the value $V = -364$ cited previously used the normalization $V \cdot R^2/n^2$ [Sánchez-Lavega et al., 2014]).

Since $V < 0$, the WKB exponent $s^2 = V < 0$ forces oscillatory solutions with $s = i\alpha_\rho$ ($\alpha_\rho \neq 0$). Writing $s = \sigma + i\alpha_\rho$: $\text{Im}(s^2) = 2\sigma\alpha_\rho = 0$. When $\alpha_\rho \neq 0$ (oscillatory region): $\sigma = 0$. The elliptic ($\sigma = 0$) and hyperbolic ($\alpha_\rho = 0$) branches are disconnected: smooth jet perturbations cannot move the solution from the oscillatory branch to the evanescent branch at leading WKB order. This is a structural stability result for the Saturn selection mechanism: the $n = 6$ stationary wave exists because $V(0) < 0$, and no continuous deformation of the jet profile can remove it without first passing through $V(0) = 0$ (which requires changing the sign of $\beta_{\text{eff}}/U_{\text{max}} - n^2/R^2$, a finite perturbation). The hexagon is therefore structurally stable at leading WKB order: it cannot be removed by infinitesimal jet perturbations.

5.4 The continuous-to-discrete bridge

The results of Sections 3–8 are stated for point vortices. Two distinct bridges connect these to continuous vorticity distributions, applicable in different regimes.

5.4.1 The vortex blob bridge (Jupiter regime)

For discrete cyclone configurations ($\beta R^3/\kappa \ll 1$), replace each point vortex with a smooth blob of width ε . The regularized interaction $h_\varepsilon(d) = -\frac{1}{2} \ln(d^2 + 2\varepsilon^2)$ recovers $-\ln d$ as $\varepsilon \rightarrow 0$.

Proposition 10 (Blob stability correction). *The constrained Lagrangian eigenvalues for h_ε satisfy*

$$\lambda_m(\varepsilon) = \lambda_m(0) + \left[-\frac{N^2-1}{12} + P_m\right] \varepsilon^2 + O(\varepsilon^4), \quad (28)$$

where $\lambda_m(0)$ is the Havelock eigenvalue (Theorem 5).

Proof. Lagrange multiplier shift. The stationarity condition $\partial H_\varepsilon / \partial x_0 = \mu_L \cdot 2x_0$ gives $\mu_L(\varepsilon) = \mu_L(0) + \delta\mu_L \cdot \varepsilon^2$ where $\delta\mu_L = (N^2-1)/24$, using $\sum_{k=1}^{N-1} \csc^2(\pi k/N) = (N^2-1)/3$. The shift $-2\delta\mu_L = -(N^2-1)/12$ is *universal* (the same for all Fourier modes).

Hessian correction. The $O(\varepsilon^2)$ corrections to h'' and h'/d are: $\delta(h''(d)) = -6\varepsilon^2/d^4 + O(\varepsilon^4)$, $\delta(h'(d)/d) = 2\varepsilon^2/d^4 + O(\varepsilon^4)$. For a pair at distance $d_p = 2 \sin(\pi p/N)$, the Hessian correction per pair is $O(\varepsilon^2/d_p^4)$. The projection $P_m = v_m^\top \delta \nabla^2 H v_m$ of this correction onto the critical eigenvector v_m is computed by summing over all $N-1$ pairwise distances with the Fourier mode weights.

Operator norm bound. Since $d_{\min} = 2 \sin(\pi/N) > 0$:

$$\|\nabla^2 H_\varepsilon - \nabla^2 H_0\|_{\text{op}} \leq 6(2N-3) \varepsilon^2 / d_{\min}^4 = O(\varepsilon^2). \quad (29)$$

By Weyl's perturbation theorem, each eigenvalue shifts by at most this amount, giving uniform convergence at $O(\varepsilon^2)$.

Profile independence. For a blob with finite second moment σ^2 (Gaussian, compact support), the Fourier transform $|\hat{\phi}(\varepsilon k)|^2 = 1 - \sigma^2 \varepsilon^2 |k|^2 + O(\varepsilon^4 |k|^4)$ shows the $O(\varepsilon^2)$ correction is a constant in k -space (= self-energy, not affecting pairwise eigenvalues). The pairwise correction is $O(\varepsilon^4)$ —*faster* than the Cauchy blob. Since the Cauchy blob is the slowest-converging regularization, all profiles with finite σ^2 inherit its stability conclusions. $\square$

The Hessian projection P_m and the total correction $c_m = -(N^2-1)/12 + P_m$ (specific to the Cauchy blob $h_\varepsilon = -\frac{1}{2} \ln(d^2 + 2\varepsilon^2)$) are exact rational numbers, computed as follows.

Differentiating the Cauchy blob pair Hessian with respect to ε^2 at $\varepsilon = 0$ gives the per-pair correction:

$$\delta H_{jj,xx} = \frac{2}{d_{jk}^6}((\Delta y)^2 - 3(\Delta x)^2), \quad \delta H_{jk,xx} = -\delta H_{jj,xx}, \quad (30)$$

with analogous expressions for yy and xy (see proof of Proposition 10). Here $(\Delta x, \Delta y, d_{jk})$ are the pair separation components. The critical radial Fourier mode is $v_k \propto z_k \cos(2\pi mk/N)$ with $m = \lfloor N/2 \rfloor$. The projection

$$P_m = v_m^\top \delta \nabla^2 H v_m \quad (31)$$

is a finite quadratic form: a sum over $N(N-1)$ pair contributions, each involving trigonometric values at N -th roots of unity. Since $\sin(\pi p/N)$ and $\cos(2\pi p/N)$ are algebraic for integer p and N , every term is rational and so is the sum. Explicit values (verified to machine precision in `blob_correction.compute_Pm(N)` and notebook `05.BlobCorrection.ipynb`):

N	3	4	5	6	7	8
$-(N^2-1)/12$	$-\frac{2}{3}$	$-\frac{5}{4}$	-2	$-\frac{35}{12}$	-4	$-\frac{21}{4}$
P_m	$-\frac{1}{3}$	$+\frac{1}{4}$	$+1$	$+4$	$+8$	$+\frac{65}{4}$
c_m	-1	-1	-1	$+\frac{13}{12}$	$+4$	$+11$

The Hessian projection P_m changes sign between $N = 3$ ($P_3 = -1/3$) and $N = 4$ ($P_4 = +1/4$): for the triangle, the blob softening reduces the nearest-neighbor curvature (all pairs are at the same short distance $d = \sqrt{3}$), while for the square and larger polygons, the long-range pairs increasingly dominate the projection and $P_m > 0$. Despite this sign change, $c_m = -1$ universally for $N \leq 5$ because the Lagrange multiplier shift $-(N^2-1)/12$ dominates P_m in this range. For $N \geq 6$, P_m grows faster and overwhelms the shift.

For blob profiles with finite second moment σ^2 (Gaussian, compact support), the $O(\varepsilon^2)$ correction is confined to the self-energy and does not affect pairwise eigenvalues (Section 5.4, profile independence argument). The pairwise correction is $O(\varepsilon^4)$ with a profile-dependent coefficient. Since the $O(\varepsilon^2)$ corrections from the Cauchy blob table above preserve stability for $N \leq 7$, and all finite- σ^2 blobs have *smaller* corrections at any given ε , the stability conclusions hold *a fortiori* for all smooth profiles.

Three consequences follow:

(i) *Stability preserved.* For $N \leq 5$: $\lambda_0 \geq 1$ and $c = -1$, so $\lambda(\varepsilon) = \lambda_0 - \varepsilon^2 > 0$ for $\varepsilon < 1$. For $N = 6$: $c > 0$, so $\lambda(\varepsilon) > \lambda_0$ —the blob *enhances* stability.

(ii) *Marginal mode resolved.* At $N = 7$, $\lambda_0 = 0$ and $c = +4$, so $\lambda(\varepsilon) = 4\varepsilon^2 > 0$ for any $\varepsilon > 0$. The heptagonal marginal stability is an artifact of the singular point vortex limit.

(iii) *Instability partially compensated.* For $N = 8$: $\lambda_0 = -1$ and $c = 11$, so the octagon is stabilized for $\varepsilon > 1/\sqrt{11} \approx 0.30$.

5.4.2 Full infinite-dimensional convergence

Proposition 10 addresses center-position perturbations of N smooth blobs. The full vorticity perturbation space is infinite-dimensional, including blob *shape* deformations. A complete stability analysis requires the Hessian of the 2D Euler energy $E[\omega] = \frac{1}{2} \iint \omega(x) G(x, y) \omega(y) dx dy$ at the blob configuration $\omega_\varepsilon = \sum_k \kappa \phi_\varepsilon(\cdot - z_k)$, decomposed into three sectors:

(a) *Center translations.* The Hessian restricted to the $2N$ -dimensional subspace of center displacements is *exactly* the blob Hessian of Proposition 10. Convergence to the point vortex Hessian at $O(\varepsilon^\alpha)$ follows from the convolution estimates.

(b) *Shape perturbations.* Mean-zero, center-preserving deformations of each blob's profile enter through the self-energy $E_{\text{self}} = \iint \phi_\varepsilon(x) G(x, y) \phi_\varepsilon(y) dx dy$. The second variation at mean-zero $\delta\omega$ is, by Plancherel and the Fourier representation $\hat{G}(\xi) = 1/(4\pi^2|\xi|^2)$ of the 2D log-Green's function:

$$\delta^2 E_{\text{self}} = \iint \delta\omega(x) G(x, y) \delta\omega(y) dx dy = \int_{\mathbb{R}^2} \frac{|\hat{\delta\omega}(\xi)|^2}{4\pi^2|\xi|^2} d\xi = \frac{1}{4\pi^2} \|\delta\omega\|_{H^{-1}}^2. \quad (32)$$

Since $\hat{\delta\omega}(0) = \int \delta\omega = 0$ (mean-zero), the integrand is zero only on a set of measure zero, so $\delta^2 E_{\text{self}} > 0$ for all non-zero mean-zero $\delta\omega$: the self-energy Hessian is positive definite in H^{-1} . The rescaling $\phi_\varepsilon(x) = \varepsilon^{-2} \phi(x/\varepsilon)$ gives $\hat{\phi}_\varepsilon(\xi) = \hat{\phi}(\varepsilon\xi)$, so

$$E_{\text{self}}[\phi_\varepsilon] = \int \frac{|\hat{\phi}(\varepsilon\xi)|^2}{4\pi^2|\xi|^2} d\xi = \varepsilon^{-2} \int \frac{|\hat{\phi}(\eta)|^2}{4\pi^2|\eta|^2} d\eta = \frac{c}{\varepsilon^2},$$

and the smallest shape eigenvalue scales as c/ε^2 . Shape modes are *unconditionally stable*.

(c) *Center-shape coupling.* The cross terms involve overlap integrals

$$C_{jk} = \iint \nabla \phi_\varepsilon(x - z_j) \cdot \nabla G(x, y) \delta\psi(y - z_k) dx dy$$

between non-adjacent blobs $j \neq k$ ($|z_j - z_k| \geq d_{\min} \gg \varepsilon$), where $\delta\psi$ is a mean-zero, center-preserving shape perturbation. Since both the monopole ($\int \delta\psi = 0$) and dipole ($\int y \delta\psi = 0$) moments of $\delta\psi$ vanish, the leading term in the multipole expansion of $G(x, y)$ around z_k is the quadrupole:

$$G(x, y) = G(x, z_k) + \nabla_y G(x, z_k) \cdot (y - z_k) + O(|y - z_k|^2/d^2), \quad d = |x - z_k|.$$

The first two terms contribute zero (by the vanishing moments), so the coupling is determined by the quadrupole interaction: $C_{jk} = O(\varepsilon^2/d_{\min}^3)$. (Numerically verified: $C_{jk} \approx 4\varepsilon^2/d_{\min}^3$ for the Cauchy blob at $d_{\min} \geq 3\varepsilon$.) This is *algebraically small*—the correct bound for the Cauchy blob, which has algebraic not exponential spatial decay. The ratio of coupling to shape eigenvalue is $O(\varepsilon^4/d_{\min}^3) \rightarrow 0$ as $\varepsilon \rightarrow 0$ with $d_{\min}$ fixed, which is the only bound needed below.

Proposition 11 (Inertia preservation). *For sufficiently small $\varepsilon > 0$, the full Hessian of $E[\omega_\varepsilon] + \mu_L L[\omega_\varepsilon]$ on the space of $\mathbb{Z}_N$ -symmetric vorticity perturbations has the same inertia (number of positive, zero, and negative eigenvalues along the constrained directions) as the point vortex Lagrangian Hessian of Theorem 5. The point vortex model is therefore the exact limit of smooth vorticity dynamics for stability analysis: no eigenvalue is lost, no sign is changed, and no new instability arises from the infinite-dimensional shape degrees of freedom. The stability boundary $N \leq 7$ is not an approximation of the smooth result—it is the smooth result in the $\varepsilon \rightarrow 0$ limit.*

Proof. Write the constrained full Hessian in block form:

$$\mathcal{H} = \begin{pmatrix} H_{cc} & H_{cs} \\ H_{cs}^\top & H_{ss} \end{pmatrix},$$

where the subscripts c and s denote the center and shape sectors.

Shape sector. By (32), $H_{ss} \geq (c/\varepsilon^2) \text{Id}_{H^{-1}}$ with $c > 0$. All shape eigenvalues are bounded below by $c/\varepsilon^2 > 0$.

Center sector. By (a) and the convolution estimates of Proposition 10, the center-sector eigenvalues satisfy $\lambda_c^{(k)}(\varepsilon) = \lambda_k^{\text{Hav}} + O(\varepsilon^2)$, where λ_k^{Hav} are the Havelock eigenvalues.

Coupling bound. From (c), the off-diagonal block satisfies $\|H_{cs}\|_{\text{op}} = O(\varepsilon^2/d_{\min}^3)$. By the Schur complement, the full Hessian has the same inertia as $H_{cc} - H_{cs}H_{ss}^{-1}H_{cs}^\top$. The correction term is

$$\|H_{cs}H_{ss}^{-1}H_{cs}^\top\|_{\text{op}} \leq \|H_{cs}\|_{\text{op}}^2 \|H_{ss}^{-1}\|_{\text{op}} = O\left(\frac{\varepsilon^4}{d_{\min}^6}\right) \cdot O(\varepsilon^2) = O\left(\frac{\varepsilon^6}{d_{\min}^6}\right),$$

which goes to zero as $\varepsilon \rightarrow 0$ with $d_{\min}$ fixed. Hence, for sufficiently small ε , the Schur complement eigenvalues are within $O(\varepsilon^6/d_{\min}^6)$ of the center-sector eigenvalues H_{cc} , and in particular preserve their signs. Since $\|H_{cc} - H_{cc}^{\text{Hav}}\|_{\text{op}} = O(\varepsilon^2)$ (from the blob correction), for small enough ε the Schur-complement eigenvalues have the same signs as the Havelock eigenvalues. Therefore the full Hessian inertia equals the Havelock inertia—which is the point vortex Lagrangian inertia of Theorem 5. $\square$

Proposition 11 assumes a prescribed blob profile ϕ_ε . For a *general* $\mathbb{Z}_N$ -symmetric vorticity ω concentrated in N patches near the N -gon vertices, the profile is not prescribed. We show the same inertia result holds for any such configuration.

Center extraction and dipole cancellation. Decompose $\omega = \sum_k \omega_k$ via Voronoi cells and define the centroid of each patch:

$$\bar{z}_k = \frac{1}{\kappa} \int x \omega_k(x) dx.$$

Any perturbation $\delta\omega_k$ decomposes as $\delta\omega_k = \delta\omega_k^{(c)} + \delta\omega_k^{(s)}$, where $\delta\omega_k^{(c)} = -\delta\bar{z}_k \cdot \nabla\omega_k$ is the center-translation component and the residual $\delta\omega_k^{(s)}$ is center-preserving: $\int x \delta\omega_k^{(s)} dx = 0$. *This is not an assumption but a consequence of the construction:*

$$\begin{aligned} \int x \delta\omega_k^{(s)} dx &= \int x \delta\omega_k dx - \int x (-\delta\bar{z}_k \cdot \nabla\omega_k) dx \\ &= \kappa \delta\bar{z}_k + \delta\bar{z}_k \cdot \int x \nabla\omega_k dx. \end{aligned}$$

Integration by parts gives $\int x_i (\partial_j \omega_k) dx = -\int \delta_{ij} \omega_k dx = -\kappa \delta_{ij}$ (the boundary term vanishes provided ω_k is compactly supported or decays faster than $|x|^{-2}$ at infinity), so $\int x \nabla\omega_k dx = -\kappa \text{Id}$, and the right-hand side is $\kappa \delta\bar{z}_k - \kappa \delta\bar{z}_k = 0$. The dipole moment of $\delta\omega_k^{(s)}$ vanishes for any profile $\omega_k \in L^1(\mathbb{R}^2)$ with finite first moment $\int |x| |\omega_k| dx < \infty$.

Coupling bound for general $\omega_k \in L^1$. The center-shape coupling C_{jk} involves the interaction between a center translation of patch j and a shape perturbation $\delta\omega_k^{(s)}$ of patch k ($j \neq k$). Expanding the Green's function $G(x, y) = -\frac{1}{2\pi} \ln|x-y|$ around the centroid $\bar{z}_k$:

$$\begin{aligned} G(x, y) &= G(x, \bar{z}_k) + \nabla_y G(x, \bar{z}_k) \cdot (y - \bar{z}_k) \\ &\quad + \frac{1}{2} (y - \bar{z}_k)^\top \nabla_y^2 G(x, \bar{z}_k) (y - \bar{z}_k) + \cdots \end{aligned}$$

Integrating against $\delta\omega_k^{(s)}(y) dy$: the zeroth-order term vanishes ($\int \delta\omega_k^{(s)} = 0$, mean-zero); the first-order (dipole) term vanishes ($\int (y - \bar{z}_k) \delta\omega_k^{(s)} = 0$, by the dipole cancellation proved above). The leading contribution is therefore the quadrupole:

$$C_{jk} = \frac{1}{2} \int \int \nabla \phi_j(x) \cdot [(y - \bar{z}_k)^\top \nabla_y^2 G(x, \bar{z}_k) (y - \bar{z}_k)] \delta\omega_k^{(s)}(y) dy dx.$$

The second derivatives $\nabla_y^2 G(x, \bar{z}_k) = O(1/d_{jk}^2)$ for $|x - \bar{z}_k| \geq d_{jk}$, and $|y - \bar{z}_k| \leq \varepsilon$ within patch k . The integral over y contributes $\int |y - \bar{z}_k|^2 |\delta\omega_k^{(s)}| dy \leq \varepsilon^2 \|\delta\omega_k^{(s)}\|_{L^1}$, giving

$$|C_{jk}| \leq \frac{C \varepsilon^2}{d_{jk}^2} \|\delta\omega_k^{(s)}\|_{L^1} \|\nabla\phi_j\|_{L^1} = O(\varepsilon^2/d_{\min}^2)$$

for any $\omega_k \in L^1$ with support in $B(\bar{z}_k, \varepsilon)$. (For the Cauchy blob, which has $|x|^{-3}$ tails extending beyond ε , the bound worsens to $O(\varepsilon^2/d_{\min}^3)$ because $\|\nabla\phi_j\|_{L^1}$ grows as $1/d_{\min}$; the $O(\varepsilon^2/d_{\min}^2)$ bound holds for compactly supported or exponentially decaying profiles.)

The self-energy Hessian (32) is positive definite for *any* mean-zero profile. The $\mathbb{Z}_N$ symmetry forces all patches to be congruent (up to rotation), so the inter-patch Hessian depends on the profile only through its convolution with G —the same structure as Proposition 10.

Therefore Proposition 11 extends to arbitrary $\mathbb{Z}_N$ -symmetric vorticity patches satisfying the L^1 regularity condition above. The phrase *sufficiently concentrated* is made quantitative as follows. The operator norm bound (29) is derived for the Cauchy blob, which has algebraic ($|x|^{-3}$) rather than exponential spatial decay and is therefore the slowest-converging regularization among profiles with finite second moment; profiles with faster decay (Gaussian, compact support) satisfy the same bound *a fortiori*, with a smaller implicit constant. The Cauchy-blob bound gives $\|\nabla^2 H_\varepsilon - \nabla^2 H_0\|_{\text{op}} \leq 6(2N-3) \varepsilon^2/d_{\min}^4$. Sign preservation of the Havelock eigenvalues requires this to be less than $\lambda_{\min}(N)$ (the smallest positive Havelock eigenvalue at the configuration), yielding the explicit concentration condition

$$\frac{\varepsilon}{d_{\min}} < \delta(N) := \sqrt{\frac{\lambda_{\min}(N)}{6(2N-3)}}, \quad (33)$$

where $d_{\min} = 2 \sin(\pi/N)$ is the nearest-neighbor separation (unit ring). Numerically:

N	4	5	6	7
$\lambda_{\min}(N)$	1	1	$\frac{1}{2}$	0
$\delta(N)$	0.18	0.15	0.096	—

Corollary 12 (Inertia preservation for $N \leq 6$). *For $N \leq 6$ and any $\mathbb{Z}_N$ -symmetric vorticity distribution with patch radius $\varepsilon/d_{\min} < \delta(N)$ (equation (33)), the full Hessian inertia equals the Havelock inertia. The argument uses only the spectral-gap condition $\lambda_{\min}(N) > 0$ and the Weyl perturbation bound; no information about the $O(\varepsilon^2)$ blob correction is needed.*

Corollary 13 (Marginal-mode resolution for $N = 7$). *For $N = 7$, the concentration bound (33) does not apply ($\lambda_{\min} = 0$, so $\delta(7) = 0$). A separate, logically distinct argument is required. The blob correction $c_7 = +4 > 0$ (Proposition 10) lifts the marginal eigenvalue to $+4\varepsilon^2$ for any $\varepsilon > 0$. The Schur-complement coupling is $O(\varepsilon^6/d_{\min}^6) \ll 4\varepsilon^2$ for all $\varepsilon < d_{\min}$, so the lifted eigenvalue remains positive. The conclusion holds for any $\mathbb{Z}_7$ -symmetric vorticity with non-overlapping patches ($\varepsilon < d_{\min}$), a weaker condition than Corollary 12. The $N = 7$ result is correspondingly more delicate: it holds at $O(\varepsilon^2)$ order and could in principle be affected by $O(\varepsilon^4)$ corrections at large $\varepsilon/d_{\min}$.*

The two corollaries rest on different mechanisms: Corollary 12 on a spectral gap, Corollary 13 on a blob correction. Both extend beyond prescribed-profile blobs to general $\mathbb{Z}_N$ -symmetric vorticity patches satisfying their respective conditions.

5.4.3 The Rossby bridge (Saturn regime)

For continuous jet meanders ($\beta R^3/\kappa \gg 1$), the bridge is not vortex concentration but *Rossby wave quantization*. The QGPV linearized about a zonal jet $U(y)$ gives the Rayleigh–Kuo equation:

$$(U - c)(\hat{\psi}'' - k^2\hat{\psi}) + (\beta - U'')\hat{\psi} = 0.$$

For a stationary wave ($c = 0$) at the jet center ($y = 0$) of a Gaussian jet with $U''(0) = -U_{\max}/\sigma^2$:

$$\hat{\psi}'' + \left[\frac{\beta_{\text{eff}}}{U_{\max}} - \frac{n^2}{R^2} \right] \hat{\psi} = 0, \quad \beta_{\text{eff}} = \beta + \frac{U_{\max}}{\sigma^2},$$

where the effective PV gradient β_{eff} includes the jet curvature (which dominates β by a factor ~ 13 for Saturn). The stationary azimuthal wavenumber is

$$n^* = R_S \cos \varphi \sqrt{\beta_{\text{eff}}/U_{\max}}, \quad (34)$$

where $R_S \cos \varphi$ is the ring radius at latitude φ ($\approx 1.31 \times 10^7$ m for Saturn at 76° N). The nearest integer $n = \text{round}(n^*)$ is the selected polygon wavenumber. For Saturn: $\beta_{\text{eff}}/U_{\max} \approx 1.72 \times 10^{-13} \text{ m}^{-2}$, giving $n^* \approx 5.4$ from the leading-order dispersion relation. The observed $n = 6$ is obtained in numerical eigenvalue computations of the full Rayleigh–Kuo equation [Sánchez-Lavega et al., 2014]; the discrepancy between the leading-order $n^* \approx 5.4$ and the observed $n = 6$ reflects corrections from the finite jet width (the meridional eigenfunction structure contributes an $O(1/\sigma^2)$ correction to the effective wavenumber that is not captured by the dispersion relation alone).

The discretization mechanism is *azimuthal periodicity*: the wave must fit an integer number of wavelengths around the ring.

The connection to the point vortex energy landscape proceeds through the *critical layer* of the saturated wave. At finite amplitude, the stationary Rossby wave develops a cat’s-eye structure at the critical line $U = c = 0$: vorticity is trapped into N lobes of width $\sim \sigma$ centered at the meander peaks. As the jet sharpens ($\sigma/R \rightarrow 0$ at fixed wave amplitude), these lobes concentrate into N isolated patches near the N -gon vertices.

Remark 7 (Cat’s-eye Hessian convergence). *The cat’s-eye vorticity decomposes as $\omega = \omega_{\text{lobe}} + \omega_{\text{braid}} + \omega_{\text{bg}}$, where the lobes are the trapped fluid inside the separatrices, the braids are thin filaments along the separatrix itself, and the background is the passing fluid outside the cat’s-eye.*

Two conditions are required for Proposition 11 to apply:

- (i) **Lobe concentration.** *The lobes satisfy the concentration condition (33) with $\varepsilon \sim \sigma$ and the centroids converge to the N -gon vertices. This is guaranteed by $\mathbb{Z}_N$ symmetry.*
- (ii) **Braid smallness.** *The absolute braid circulation Γ_{braid} is negligible compared to the total lobe circulation Γ_{lobe} .*

Both conditions are verified numerically on the Stuart vortex $\Psi = -\sigma^2 \ln(\cosh(y/\sigma) - \eta \cos(n\theta))$, an exact steady Euler solution with analytical cat’s-eye structure, where $\eta \in (0, 1)$ is the Stuart amplitude parameter ($\eta = 0$: linear wave; $\eta \rightarrow 1$: point-vortex concentration—note this is the opposite convention from the blob width ε in Section 5.4, where $\varepsilon \rightarrow 0$ is the point-vortex limit). As $\eta \rightarrow 1$:

- The lobe fraction of total circulation approaches unity: $\Gamma_{\text{lobe}}/\Gamma_{\text{total}} = 1 - O(1-\eta)$, with $\Gamma_{\text{braid}}/\Gamma_{\text{total}} = O(1-\eta)$ (fitted exponent 1.08, $R^2 = 0.997$).
- At $\eta = 0.99$, the lobes capture 99.6% of the total circulation; braids carry 0.1%.
- The $\mathbb{Z}_N$ -symmetric lobe circulations are equal to within 0.01%, and the centroids lie on the N -gon vertices (both by the $\mathbb{Z}_N$ symmetry of the cat's-eye).
- The constrained Hessian of the lobe circulations placed on the N -gon ring has the same inertia (number of positive, zero, and negative eigenvalues) as the Havelock Hessian, for all $\eta \in [0.3, 0.99]$.

The same inertia result holds for the Prandtl–Batchelor cat's-eye (homogenized PV inside the lobes, discontinuous at the separatrix), which is the physically relevant model for Saturn's long-lived hexagon. Code: `catseye_decomposition.py`, `catseye_hessian.py`.

The Saturn bridge thus arrives at the *same* energy landscape as the Jupiter bridge, through a different limiting procedure: Saturn concentrates a continuous jet meander into N patches (jet width $\sigma \rightarrow 0$), while Jupiter concentrates N discrete cyclones into point vortices (blob width $\varepsilon \rightarrow 0$). The two limits share the same Hessian inertia because both preserve the $\mathbb{Z}_N$ symmetry that forces equal lobe circulations and N -gon geometry.

5.4.4 Resilience to circulation disorder

In both planetary regimes the vortex circulations κ_k are not perfectly equal; convection injects or removes circulation from individual vortices continuously. We model this as $\kappa_k = \kappa(1 + \eta_k)$ with i.i.d. fractional disorders $\eta_k \sim \mathcal{N}(0, \sigma_\eta^2)$.

We work on $\mathbf{H}^2$ rather than the flat plane because the hyperbolic setting simultaneously tests two effects: circulation disorder and curvature stabilization. In the flat limit $\rho/a \rightarrow 0$ the $\mathbf{H}^2$ eigenvalues recover the Havelock values exactly, so the flat-plane result is contained as a special case; conversely, the curvature-stabilized configurations ($N \geq 8$ on $\mathbf{H}^2$) have no flat-plane analog and can only be probed on the curved surface.

On $\mathbf{H}^2$, the disordered Hamiltonian is

$$H_\kappa = - \sum_{j < k} \kappa_j \kappa_k \ln |z_j - z_k| + \frac{1}{2} \sum_j \kappa_j (K_{\text{tot}} - \kappa_j) \ln(a^2 - |z_j|^2), \quad (35)$$

where $K_{\text{tot}} = \sum_k \kappa_k$. At $\eta_k \equiv 0$ this reduces exactly to the uniform Hamiltonian H_{hyp} .

Holding the ring positions at the uniform equilibrium (exact to $O(\eta)$; the $O(\eta)$ shift of equilibrium produces only an $O(\eta^2)$ eigenvalue correction), we compute the minimum Fourier-mode eigenvalue $\lambda_{\min}(\eta)$ via finite differences. Monte Carlo over $n = 1000$ realizations estimates $P(\lambda_{\min} < 0)$ as a function of (σ_η, N, ρ) .

Physical range of σ_η . Juno imagery [Adriani et al., 2018, Gavriel and Kaspi, 2021] shows Jupiter's eight north-polar cyclones have peak wind speeds ranging from ~ 60 to $\sim 100 \text{ m s}^{-1}$ and diameters from $\sim 4,000$ to $\sim 7,000 \text{ km}$. Taking circulation $\kappa \propto V_{\text{max}} r_c$, the coefficient of variation across the eight cyclones is $\sigma_\eta \approx 0.15\text{--}0.25$. Saturn's hexagonal jet is more uniform (Cassini wind profiles give $\sigma_\eta \lesssim 0.05$ for the azimuthal circulation variation). The physically relevant range is therefore $\sigma_\eta \in [0.05, 0.25]$.

Key findings (all uncertainties are $\pm 1\sigma$ binomial standard errors $\sqrt{P(1-P)/n}$ with $n = 1000$):

- **Stable configurations are robust.** For $N = 6$ at $\rho = 0.5a$, the instability probability remains below 10% ($\pm 0.9\%$) for $\sigma_\eta \leq 0.2$, covering the full physically relevant range.
- **Curvature-stabilized rings ($N \geq 8$) are similarly robust.** At $N = 8$, $\rho = 2a$ (well inside the stable region on $\mathbf{H}^2$), $P < 0.5$ ($\pm 1.6\%$) for $\sigma_\eta \leq 0.1$. At $\sigma_\eta = 0.2$ (Jupiter’s estimated value), P remains below 15%.
- **The $N = 7$ marginal ring is sensitive.** At $N = 7$, $\rho = 0.05a$ (flat-limit regime), $P(\lambda_{\min} < 0)$ increases monotonically with σ_η : $P \approx 0$ ($\pm 0.3\%$) at $\sigma_\eta = 0.1$ and $P > 0$ for $\sigma_\eta \geq 0.5$. The marginal mode ($\lambda_0 = 0$ at $K = 0$) is the first to be destabilized by disorder, as expected. At Jupiter’s estimated $\sigma_\eta \approx 0.2$, a hypothetical $N = 7$ ring would have $P \approx 0$ —the marginal sensitivity manifests only at unphysically large disorder.

Thus the N_{crit} boundary is a *stability domain boundary*, not a *knife-edge*: configurations comfortably inside the stable region remain stable under the circulation fluctuations observed at Jupiter ($\sigma_\eta \approx 0.2$) and Saturn ($\sigma_\eta \lesssim 0.05$).

Code: `circulation_disorder.instability_probability(N, ρ, ση)` and `disorder_min_eigenvalue` in `src/planetary_polygons/extensions/circulation_disorder.py`; test suite in `tests/test_circu`

5.4.5 The regime criterion

The dimensionless number $\beta R^3/\kappa$ is the ratio of two timescales: the Rossby wave period $T_R \sim 1/(\beta R)$ and the vortex turnover time $T_v \sim R^2/\kappa$. When $\beta R^3/\kappa = T_v/T_R \ll 1$, the vortex interactions complete many orbits before the Rossby wave propagates appreciably, so the dynamics is vortex-dominated. When $\beta R^3/\kappa \gg 1$, the Rossby wave dynamics operates faster than the vortex rearrangement, so the wave mechanism dominates:

Regime	$\beta R^3/\kappa$	Bridge
Vortex-dominated	$\ll 1$	Blob convergence (Proposition 10)
Wave-dominated	$\gg 1$	Rossby stationarity (equation (34))

Jupiter’s north pole. Throughout this regime calculation κ denotes the *Rankine-core* circulation, which integrates only within the solid-body rotation core and sets the characteristic vortex timescale $T_v \sim R^2/\kappa$. The full-cyclone circulation κ_{full} (which includes the outer irrotational skirt) is larger and is the appropriate quantity for computing orbital periods and eigenfrequencies; it is used there instead (Section 8.3). The two estimates are numerically distinguished below.

At 84°N , the planetary vorticity gradient is $\beta_J = 2\Omega_J \cos(84^\circ)/R_J \approx 5.1 \times 10^{-13} \text{ m}^{-1}\text{s}^{-1}$ and the cyclone-ring radius is $R \approx 7,470 \text{ km}$. Each Juno-observed circumpolar cyclone has Rankine core radius $r_c \approx 700 \text{ km}$ and peak wind $V_{\text{max}} \approx 80 \text{ m s}^{-1}$ [Adriani et al., 2018, Gavriel and Kaspi, 2021], giving $\kappa_{\text{Rank}} = 2\pi r_c V_{\text{max}} \approx 3.5 \times 10^8 \text{ m}^2 \text{ s}^{-1}$ (full-cyclone estimate: $r_c \approx 3,200 \text{ km}$, $\kappa_{\text{full}} \approx 1.6 \times 10^9 \text{ m}^2 \text{ s}^{-1}$). Hence

$$\frac{\beta R^3}{\kappa} = \frac{5.1 \times 10^{-13} (7.47 \times 10^6)^3}{3.5 \times 10^8} \approx 0.61.$$

This places Jupiter in the vortex-dominated regime, but only moderately: 0.61 is $O(1)$, not $\ll 1$. Rossby-wave effects are not negligible at this value, and a decrease in the

effective κ (e.g. using the Rankine-core rather than full-cyclone estimate) would push the ratio above unity. The qualitative conclusion—discrete cyclones, Thomson stability applies, blob bridge proves convergence—is robust to this uncertainty because the vortex-dominated bridge requires only that the cyclones be identifiable as N distinct vortices (which the Juno imagery unambiguously confirms), not that $\beta R^3/\kappa$ be parametrically small. The ratio enters quantitatively only through the Rossby-wave correction to the eigenvalues, which is $O(\beta R^3/\kappa)$ and does not change the sign of $\lambda_{\min}$ for $N = 8$ (where $|\lambda_{\min}| = 1 \gg 0.61$). Using the full-cyclone circulation $\kappa_{\text{full}} \approx 1.6 \times 10^9 \text{ m}^2 \text{ s}^{-1}$ gives $\beta R^3/\kappa \approx 0.13$, firmly in the vortex-dominated regime.

Saturn’s hexagon. At 76°N , $\beta_S = 2\Omega_S \cos(76^\circ)/R_S \approx 1.3 \times 10^{-12} \text{ m}^{-1} \text{ s}^{-1}$ and $R \approx 14,600 \text{ km}$. Saturn’s polar jet is well-described by a Gaussian profile $U = U_{\max} \exp(-y^2/2\sigma^2)$; its effective circulation is $\kappa = U_{\max} \sqrt{2\pi} \sigma$. With $U_{\max} \approx 100 \text{ m s}^{-1}$ and $\sigma \approx 2,100 \text{ km}$ (Cassini VIMS wind profiles), $\kappa \approx 5.4 \times 10^8 \text{ m}^2 \text{ s}^{-1}$, giving

$$\frac{\beta R^3}{\kappa} = \frac{1.3 \times 10^{-12} (1.46 \times 10^7)^3}{5.4 \times 10^8} \approx 7.6.$$

Continuous jet meander, Rossby stationarity selects $n = 6$, periodicity provides quantization.

Both mechanisms require the 2D logarithmic Green’s function as the common mathematical kernel.

5.5 Finite Rossby deformation radius

In quasi-geostrophic dynamics with finite Rossby deformation radius R_d , the Green’s function is $K_0(r/R_d)$ (modified Bessel function), which recovers $-\ln r$ for $r \ll R_d$. The interaction $h(r) = K_0(r/R_d)$ satisfies $h'(r) = -K_1(r/R_d)/R_d < 0$ and $(rh'(r))' = rK_0(r/R_d)/R_d^2 > 0$ (class (iii)). Numerical computation of the constrained Lagrangian Hessian as a function of $\eta = R/R_d$ (polygon ring radius over deformation radius) reveals an N -dependent *polygon-suppression transition*:

N	Barotropic $\lambda_{\min}$	Critical R/R_d	Status
≤ 5	≥ 1	none	Stable for all R/R_d
6	$1/2$	≈ 1.29	Suppressed for $R/R_d > 1.29$
7	0	≈ 0	Unstable for any finite R/R_d
≥ 8	< 0	N/A	Unstable (barotropic)

Polygons with $N \leq 5$ are robust against stratification effects. The hexagon ($N = 6$) is the first polygon sensitive to R_d : it requires $R/R_d < 1.29$. The heptagon ($N = 7$), neutrally stable in the barotropic limit, is destabilized by any finite R/R_d —providing an additional mechanism for its rarity beyond the quartic confinement.

For Saturn, the barotropic deformation radius ($\sim 2000\text{--}3000 \text{ km}$) gives $R/R_d \approx 8\text{--}12$ at the hexagonal jet, placing it above the critical ratio. The observed hexagonal persistence therefore requires the dynamics to be effectively barotropic over a deep layer—a requirement now directly supported by observational data. Cassini Grand Finale gravity measurements constrain Saturn’s differential rotation to a depth of $8,000\text{--}9,000 \text{ km}$ [Iess et al., 2019], implying a coherent barotropic column of order 10^4 km , far exceeding the hexagon’s vertical scale. Saturn is therefore the best-supported case in our sample for the applicability of the 2D logarithmic Green’s function: the columnar depth is observationally established, not merely assumed.

Observable prediction: hexagon suppression signatures. The critical ratio $R/R_d \approx 1.29$ for $N = 6$ yields a testable prediction. If the effective barotropic depth were reduced (by seasonal radiative forcing of the upper troposphere, or by a hypothetical change in deep convective coupling), R_d would decrease and R/R_d would increase toward the critical value. Approaching the threshold, the framework predicts three observable signatures: (1) the minimum Havelock eigenvalue $\lambda_{\min}(\eta) \rightarrow 0$, reducing the hexagonal mode’s restoring frequency and producing a *longer oscillation period* for meridional meander perturbations; (2) the hexagonal amplitude ε_{obs} would *grow*, since the restoring force weakens while convective forcing remains constant; and (3) at $R/R_d > 1.29$, the hexagonal mode becomes linearly unstable and the pattern would transition to a lower- N polygon ($N = 5$, the next Thomson-stable configuration) or break up entirely. Saturn’s hexagon has been observed continuously since Voyager 1 (1980–81); Cassini monitored it through northern spring equinox (2009) to summer solstice (2017) with no sign of weakening [Antuñano et al., 2015, Sánchez-Lavega et al., 2014]. This four-decade persistence across half a Saturnian year places an empirical lower bound on the effective barotropic depth: seasonal radiative forcing does not reduce R_d enough to approach the critical ratio, consistent with the deep ($\sim 10^4$ km) barotropic column implied by Cassini gravity.

For Jupiter the situation is different, and the 2D justification requires separate examination. Juno gravity measurements [Kaspi et al., 2018] constrain Jupiter’s differential rotation to $\sim 3,000$ km depth ($\approx 4\%$ of R_J), far shallower than Saturn’s. Jupiter’s polar cyclones therefore do *not* rest on deep barotropic Taylor columns; the relevant 2D dynamics is that of a shallow weather layer above a near-solid-body rotating deep interior. Two limiting cases bracket the problem:

Deep weather-layer limit ($H_{\text{eff}} \sim 3,000$ km). Taking the full differential-rotation depth as the effective layer and using the equivalent-barotropic deformation radius $R_d = \sqrt{g H_{\text{eff}}}/f \approx \sqrt{25 \times 3 \times 10^6} / (2\Omega_J \sin 84^\circ) \approx 25,000$ km, we obtain $R/R_d \approx 0.3 \ll 1$. In this limit $K_0(r/R_d) \rightarrow -\ln r + \text{const}$, so the logarithmic Green’s function applies—though the physical justification is a shallow equivalent-barotropic layer rather than a deep Taylor column.

Tropospheric limit ($H_{\text{eff}} \sim 30\text{--}100$ km). If the active polar-cyclone layer is shallower, reduced gravity $g' \sim 0.1\text{--}1$ m s $^{-2}$ gives $R_d \sim 2,000\text{--}6,000$ km and $R/R_d \sim 1\text{--}4$. The $N = 8$ ring lies in the finite- R_d regime; whether it remains stable under the K_0 interaction with central vortex $\kappa_0/\kappa \approx 0.7$ depends on the κ_0 -shifted critical ratio (not tabulated here, but the sign argument of Section 3 continues to hold since $K'_0 < 0$).

In either case the logarithmic interaction is the appropriate leading-order approximation. The effective layer depth is not directly constrained by current Juno data, making Jupiter the *least* observationally supported case for the 2D assumption, in contrast to Saturn. A direct measurement of the depth of the polar cyclone roots—analogueous to the gravity constraints on the jets—would resolve this ambiguity.

The critical ratio is defined implicitly by the equation $\bar{\lambda}(\eta_{\text{crit}}, N) = 0$, where the dimensionless eigenvalue $\bar{\lambda}(\eta, N) = R^2 \cdot \lambda_{\min}(\eta)$ interpolates smoothly from the Havelock value $\bar{\lambda}(0, N) = (N-1) - \lfloor N/2 \rfloor (N - \lfloor N/2 \rfloor)/2$ to zero.

The Lagrange multiplier is determined by the stationarity condition $\partial H/\partial R + \mu_L \partial L/\partial R = 0$. For $H = -\sum_{j < k} K_0(d_{jk}/R_d)$ with $d_{jk} = 2R \sin(\pi p/N)$ and $L = NR^2$, the radial derivative of each pair contributes $-K'_0(d_{jk}/R_d) \cdot 2 \sin(\pi p/N)/R_d = K_1(d_{jk}/R_d) \cdot 2 \sin(\pi p/N)/R_d$ (using $K'_0 = -K_1$). Projecting onto the constraint gradient $\partial L/\partial R =$

$2NR$ and solving gives

$$-2\mu_L = \frac{\eta}{R} \sum_{p=1}^{N-1} \frac{(1 - \cos \theta_p) K_1(2\eta \sin(\pi p/N))}{2 \sin(\pi p/N)}, \quad \theta_p = 2\pi p/N, \quad (36)$$

where $\eta = R/R_d$ and the factor $(1 - \cos \theta_p)$ arises from the Fourier decomposition of the pairwise Hessian along the radial mode. As $\eta \rightarrow 0$: $K_1(x) \sim 1/x$ for small x , so each summand reduces to $(1 - \cos \theta_p)/(4\eta \sin^2(\pi p/N)) \cdot (\eta/R) = (1 - \cos \theta_p)/(4R \sin^2(\pi p/N))$. Summing over p using the Havelock identity $\sum_p (1 - \cos \theta_p)/(4 \sin^2(\pi p/N)) = (N-1)/2$ recovers the flat-plane value $-2\mu_L = (N-1)/(2R^2)$. For $N = 6$, the critical mode $m = 3$ has zero Havelock weight at $p = 2, 4$ (since $\cos(2\pi \cdot 2 \cdot 3/6) = 1$), so the critical eigenvalue depends on K_0 and K_1 at only *two* distinct arguments, η and 2η , rather than three.

No closed-form analog of the Havelock identity is known for K_0 . The orthogonal-ity mechanism that collapses the logarithmic sum to $m(N-m)/2$ relies on the identity $\sum_p e^{2\pi i m p/N} \ln |2 \sin(\pi p/N)| = 0$, which has no counterpart for K_0 : the Bessel values $K_0(2\eta \sin(\pi p/N))$ are transcendental at nonzero algebraic arguments by Siegel's theorem on E -functions, but this does not preclude the full weighted sum from having a closed form. Whether $\sum_p (\cos \theta_{pm}) K_0(2\eta \sin(\pi p/N))$ admits a closed form for any N and m is an open question; no such form has been found, and all stability computations for K_0 interactions are carried out numerically. A perturbative expansion $K_0(x) \approx -\ln x + O(x^2)$ gives $\eta_{\text{crit}} \approx 0.9$ for $N = 6$ —30% below the exact value—because the critical ratio lies in the non-perturbative regime where K_0 departs significantly from $-\ln$. The transition occurs precisely where the exponential decay of K_0 (destroying long-range correlations between non-adjacent vortices) overtakes the logarithmic regime, making the suppression inherently non-perturbative. The implicit equation is nevertheless well-posed, and the critical η is uniquely determined by bisection for each N .

The two phase boundaries. For K_0 , two distinct boundaries exist in the (N, η) plane: (1) the *spiral sign change* $\bar{H}''(0) = 0$, where the positive A_m term from $(rh')' > 0$ overwhelms the negative B_m term; and (2) the *constrained stability loss* $\bar{\lambda}_{\min} = 0$, where the Lagrange multiplier shift consumes the stability margin. The constrained boundary ($\eta \approx 1.29$ for $N = 6$) is always *inside* the spiral boundary ($\eta \approx 3.2$ for $N = 6$) because the Lagrange shift reduces the margin before H'' changes sign. Physically, the polygon requires long-range correlations between non-adjacent vortices. When $\eta \gg 1$ and the K_0 interaction decays exponentially, only nearest-neighbor correlations survive and the polygon loses its energetic advantage.

For comparison, the Cauchy-blob regularization $-\frac{1}{2} \ln(d^2 + 2\varepsilon^2)$ has $H'' < 0$ for *all* ε —the soft-core never flips the spiral sign. The Gaussian-screened interaction $-\ln(d) \cdot e^{-d^2/L^2}$ shows an earlier transition ($\eta \approx 0.9$ for $N = 6$) due to its faster-than-exponential decay.

The constrained stability boundary. The interaction class determines N_{crit} , defined here as the largest N for which the *constrained* Lagrangian Hessian $\bar{\lambda}_{\min} \geq 0$ (i.e. the polygon is a constrained energy minimum). This is distinct from the spiral-direction boundary $H''(0) = 0$, which occurs at larger η (e.g. $\eta \approx 3.2$ vs. 1.29 for $N = 6$ with K_0 ; see the two phase boundaries above):

Class	Condition	N_{crit} (constrained)
(i)	$(rh')' < 0$	≥ 7 (increases with $ (rh')' $)
(ii)	$(rh')' = 0$ (only $-\ln r$)	$= 7$ exactly
(iii)	$(rh')' > 0$ (e.g. K_0)	≤ 7 (decreases with screening)

Numerical computation confirms: for $h = -r^\alpha$ (class (i)), $N_{\text{crit}} = 7$ at $\alpha \rightarrow 0$ and increases with α ($N_{\text{crit}} = 8$ at $\alpha = 1$, $N_{\text{crit}} = 14$ at $\alpha = 2$). At $N = 7$, $\lambda_{\text{min}}(\alpha) > 0$ for all $\alpha > 0$ tested, with $d^2\lambda_{\text{min}}/d\alpha^2 = +2.5 > 0$ at $\alpha \rightarrow 0$: the marginal stability at $N = 7$ is *unique to the logarithmic interaction*, and any class (i) interaction makes the heptagon strictly stable. For $K_0(r/R_d)$ (class (iii)), the dimensionless eigenvalue $\bar{\lambda}(\eta)$ decreases monotonically in η for each Fourier mode (verified for $N = 5, 6$), consistent with the physical expectation that screening weakens the stabilizing long-range correlations. For $N_{\text{crit}} = 6$ at $R/R_d = 1$ and decreases to 5 at $R/R_d = 3$. The logarithmic interaction is the *watershed*: the critical case between interactions that stabilize polygons (class (i)) and interactions that suppress them (class (iii)).

Proposition 14 (Min/max duality). *For any pairwise interaction $h(r)$, the constrained energy minimum of $H = \sum h(d_{jk})$ and the constrained energy maximum of $-H$ have the same stability boundary N_{crit} .*

Proof. The Lagrange multiplier for $-H$ is $\mu_L^{-H} = -\mu_L^H$ (from $-\nabla H = (-\mu_L)\nabla L$). The Lagrangian Hessian $\mathcal{L}_{-H} = -\nabla^2 H + 2\mu_L I = -\mathcal{L}_H$. The eigenvalues are negated; constrained minimum (all ≥ 0) of H is equivalent to constrained maximum (all ≤ 0) of $-H$. $\square$

For gravity ($h = -1/r$, $h' > 0$), the N -gon is a constrained energy maximum. The duality gives $N_{\text{crit}}^{\text{gravity}} = N_{\text{crit}}^{\text{Coulomb}}$. Numerical computation of the Coulomb ($h = +1/r$) constrained Hessian gives $N_{\text{crit}} = 5$: the minimum eigenvalue is $+0.73$ at $N = 5$ and -0.02 at $N = 6$. This is consistent with Maxwell's (1859) result that gravitating rings require a central mass for $N \geq 7$; the one-unit difference ($N_{\text{crit}} = 5$ vs Maxwell's 6) arises because our analysis does not include the central mass (Maxwell's stability includes the stabilizing effect of the central body).

The Coulomb/gravity $N_{\text{crit}} = 5$ is less than the logarithmic $N_{\text{crit}} = 7$, confirming the class (iii) reduction: the $1/r$ interaction has $(rh')' = 1/r^2 > 0$, placing it in the destabilizing regime relative to $-\ln r$.

6 Planetary applications

6.1 Polygon quality metrics and cross-planetary comparison

The two planetary polygon types require different quality metrics. For *discrete vortex rings* (Jupiter), the natural measure is the geometric deviation from a perfect N -gon:

$$\sigma_{\text{geom}} = \frac{1}{N} \sum_{k=1}^N |\ln |z_k/z_k^{\text{ideal}}||, \quad (37)$$

where z_k^{ideal} are the best-fit regular N -gon vertices. This measures positional scatter of discrete cyclones; small σ_{geom} means the ring closely approximates the Thomson equilibrium.

For *continuous jet meanders* (Saturn), the vortex positions z_k are not defined. The appropriate measure is the fractional radial displacement of the jet center:

$$\varepsilon_{\text{obs}} = A_n/\bar{R}, \quad (38)$$

where A_n is the half-amplitude of the n -th Fourier mode of the jet-center radial position and $\bar{R}$ is the mean ring radius. For Saturn’s hexagon ($n = 6$, $\bar{R} \approx 13,200$ km), the observed $\pm 2^\circ$ latitude oscillation [Godfrey, 1988, Antuñano et al., 2015] gives $A_6 \approx 1,900$ km and $\varepsilon_{\text{obs}} \approx 0.14$ (range 0.11–0.18 across measurement epochs).

These two metrics are *not interchangeable*: applying σ_{geom} to Saturn (by treating the meander peaks as point positions) yields 0.50, an order of magnitude larger than Jupiter’s 0.05–0.06, but this comparison is physically meaningless—it conflates meander amplitude with positional scatter. Table 1 reports both metrics, using each where it is appropriate.

Table 1: Planetary polygon configurations. κ_0/κ is estimated from Juno imagery [Adriani et al., 2018] using cyclone area as a proxy ($\sim 30\%$ systematic uncertainty). “—” indicates the quantity is not applicable.

System	N	Quality metric	κ_0/κ	Selection mechanism
Saturn hexagon	6	$\varepsilon_{\text{obs}} = 0.14$	N/A (jet)	Rossby stationarity
Jupiter north	8	$\sigma_{\text{geom}} = 0.06$	~ 1	Thomson + Onsager
Jupiter south	5	$\sigma_{\text{geom}} = 0.05$	—	Thomson-stable; N not predicted here

6.2 Broader observational context

Beyond the three planetary configurations, the theoretical stability boundary $\kappa_{\text{crit}}(N)$ from Proposition 7 can be compared with laboratory and numerical observations:

Laboratory experiments. Jansson et al. [2006] and Barbosa Aguiar et al. [2010] observe polygons with $N = 2$ through 6 in rotating-tank experiments by varying the differential rotation rate. All observed N values fall in the $\kappa_{\text{crit}} = 0$ range ($N \leq 7$), consistent with the Thomson stability boundary. No laboratory experiment has produced a stable polygon with $N \geq 8$ without a central vortex—also consistent. (The laboratory mechanism is barotropic instability of a shear layer, not Thomson point vortex dynamics, so the agreement is structural rather than quantitative.)

Numerical simulations. Siegelman et al. [2022] produce $N = 8$ (north) and $N = 5$ (south) vortex crystals from 2D turbulence on polar caps, with a central polar cyclone emerging alongside the ring in both cases. The central cyclone provides the stabilization required by Proposition 7 for $N = 8$. The Siegelman et al. [2022] simulations include continuous small-scale forcing (representing convective injection) and large-scale dissipation, yet the polygon crystal is the persistent stationary state: the system is not frozen into a one-time equilibrium but continuously re-attracted to it. Within the 2D barotropic framework, this is numerical evidence that the constrained energy minimum acts as a *dynamical attractor* under turbulent forcing. The caveat (Section 5) is that these are idealized 2D simulations, not a faithful model of Jupiter’s three-dimensional polar dynamics; the agreement supports the Onsager picture but does not constitute a proof of its applicability.

Uranus. Akins et al. [2023] detect a single polar cyclone at Uranus’s north pole—a single-vortex ($N = 1$) configuration, below the polygon threshold. No ring structure has been observed. Whether the absence of a Uranian polygon reflects insufficient vortex formation (macro-scale constraint) or insufficient observations (the pole has been poorly resolved) remains open.

Neptune. No polar polygon has been observed, and Voyager 2 imagery is insufficient to resolve ring structure. However, the framework makes a falsifiable prediction.

Regime: At 84°N , $\beta_N = 2\Omega_N \cos(84^\circ)/R_N \approx 9.2 \times 10^{-13} \text{ m}^{-1}\text{s}^{-1}$ and $R_{\text{ring}} \approx 2,570 \text{ km}$, giving $\beta R^3 \approx 1.6 \times 10^7 \text{ m}^2 \text{ s}^{-1}$. For Jovian-scaled cyclones (core radius $r_c \approx 240 \text{ km}$, $V_{\text{max}} \approx 80\text{--}100 \text{ m s}^{-1}$), $\kappa \approx 1.2\text{--}1.5 \times 10^8 \text{ m}^2 \text{ s}^{-1}$ and

$$\frac{\beta R^3}{\kappa} \approx 0.10\text{--}0.13 \ll 1.$$

Neptune is firmly vortex-dominated—more so than Jupiter ($\beta R^3/\kappa \approx 0.6$), because its smaller radius ($R_N \approx R_J/3$) reduces R_{ring}^3 by a factor of ~ 25 despite a slightly larger β . The Rossby stationarity mechanism does not apply; the relevant bridge is Thomson/Onsager.

Prediction: If Neptune’s deep convection (evidenced by the Great Dark Spot and supersonic mid-latitude winds) generates polar cyclones, Onsager inverse cascade should concentrate same-sign vorticity near the poles. The Thomson stability criterion then predicts a ring of $N = N_{\text{max}}(\kappa_0/\kappa)$ cyclones at colatitude $\sim 6^\circ$ (by analogy with Jupiter’s 84°N ring), with ring radius $\approx 2,000\text{--}3,000 \text{ km}$ and $N \in \{5, 6, 7, 8\}$ depending on the polar-vortex-to-ring strength ratio κ_0/κ .

Why no specific N is predicted here: the Thomson upper bound requires knowing κ_0/κ ; the Onsager maximum- N selection then uniquely determines N given that ratio. Both are measurable from wind-field observations of Neptune’s poles—currently unconstrained. This prediction is testable by a Neptune polar orbiter (e.g. Trident-class mission).

All observations are consistent with the theoretical stability boundary: $N \leq 7$ is observed without a central vortex, $N = 8$ is observed only with one, and no $N \geq 9$ ring has been reported in any system.

6.3 Multi-ring configurations

Proposition 15 (Center-ring dichotomy). *Let N outer vortices sit at the regular N -gon on a ring of radius R_2 , and let $M \geq 2$ inner vortices sit at the regular M -gon on a ring of radius $R_1 < R_2$.*

- (i) *If $N \leq 7$ (outer ring Thomson-stable), the two-ring configuration is unstable: the Lagrangian Hessian has at least one negative eigenvalue.*
- (ii) *If $N \geq 8$ (outer ring Thomson-unstable), the inner ring can reduce but not eliminate the instability: the two-ring Hessian retains at least one negative eigenvalue.*

In contrast, a single central vortex ($M = 0$) at the $\mathbb{Z}_N$ fixed point produces a uniform eigenvalue shift $+\kappa_0/R^2$ across all Fourier modes and can achieve full stabilization (Proposition 7).

Proof. Consider the limit $R_2/R_1 \rightarrow \infty$. The angular impulse $L = MR_1^2 + NR_2^2$ is dominated by the outer ring, so the Lagrange multiplier $\mu_L \rightarrow -(N-1)/(4R_2^2) \rightarrow 0$. The inner ring’s $m=0$ Fourier mode has unconstrained eigenvalue $-(M-1)/(2R_1^2)$ (the rotation rate of the M -ring, which is the Havelock value at $m=0$ without the Lagrange shift). With $\mu_L \rightarrow 0$, this eigenvalue is not lifted, giving

$$\lambda_{\min} \rightarrow -\frac{M-1}{2R_1^2} < 0 \quad \text{for all } M \geq 2. \quad (39)$$

(Verified numerically: $\lambda_{\min} \cdot R_1^2 \rightarrow -(M-1)/2$ to six decimal places for $R_2/R_1 = 10^3$, independently of N .)

The negative eigenvalue is a continuous function of R_2/R_1 and is strictly negative at $R_2/R_1 = \infty$. Numerical computation over 108 configurations ($N \in \{5, \dots, 10\}$, $M \in \{2, \dots, 6\}$, $R_2/R_1 \in \{1.5, 2, 3, 5\}$; code in `two_ring.py`) confirms that it remains negative at all finite R_2/R_1 , establishing both parts.

The mechanism: a central vortex at the $\mathbb{Z}_N$ fixed point contributes a uniform eigenvalue shift $+\kappa_0/R^2$ across all Fourier modes, compensating the negative Havelock eigenvalues. An inner ring of $M \geq 2$ vortices shares the same Lagrange multiplier as the outer ring but has a *different* rotation rate, creating a Lagrange-multiplier mismatch (39) that produces at least one negative eigenvalue regardless of the coupling strength. $\square$

Jupiter’s observed center-plus-octagon geometry is therefore *uniquely stabilized* among all configurations with a secondary vortex structure: only the single central cyclone achieves full stabilization of the $N = 8$ ring.

7 Predictions: BEC vortex clusters

In a quasi-2D Bose–Einstein condensate in a circular trap, quantized vortices interact logarithmically at scales above the healing length ξ . The circulation is quantized: $\kappa = q \cdot h/m$ with integer winding number q . The κ_0/κ ratio of Proposition 7 becomes the winding number ratio q_0/q , yielding sharp integer-valued predictions:

Configuration	κ_{crit}	Prediction
$N \leq 7$ ring	0	Stable without central vortex
$N = 8$ ring	1/2	Unstable alone; stable with $q_0 \geq 1$
$N = 9$ ring	1	Marginally stable with $q_0 = 1$; stable with $q_0 \geq 2$
$N = 10$ ring	7/4	Unstable with $q_0 = 1$; stable with $q_0 \geq 2$

The $N = 8$ prediction is the sharpest: a ring of 8 unit-winding vortices should be unstable alone and stable upon adding a single central vortex ($q_0 = 1 > \kappa_{\text{crit}} = 1/2$).

The $N = 7$ case offers a distinct signature. The quartic restoring force ($\alpha_0 = 45/14$, Section 3.5) produces an *amplitude-dependent* oscillation frequency for the $m = 3$ mode. In the normal form $\mathcal{H} = (\alpha_0/4)\varepsilon^4$, the nonlinear oscillation frequency is $\omega_{m=3} = \alpha_0 \varepsilon^2 \kappa / (2\pi R^2)$, where ε is the dimensionless perturbation amplitude (radial displacement / ring radius) and $\kappa = h/m_{\text{atom}}$ is the single-quantum circulation. For ^{87}Rb in a circular trap of radius $R = 50 \mu\text{m}$: $\kappa \approx 1.2 \times 10^{-7} \text{ m}^2 \text{ s}^{-1}$, $\kappa / (2\pi R^2) \approx 7.6 \text{ s}^{-1}$, so $\omega_{m=3} \approx 24 \varepsilon^2 \text{ s}^{-1}$. At a 5% radial displacement ($\varepsilon = 0.05$): $\omega_{m=3} \approx 0.06 \text{ s}^{-1}$ (period $\sim 100 \text{ s}$), resolvable with standard BEC vortex imaging at $\sim 10 \text{ Hz}$ frame rates. The key signature distinguishing the $N = 7$ quartic confinement from the $N \leq 6$ quadratic regime is that the $m = 3$ oscillation frequency *vanishes* as $\varepsilon \rightarrow 0$ (rather than approaching a finite value), producing an amplitude-dependent period that is directly measurable.

The healing length ξ provides a natural blob width $\varepsilon_{\text{blob}} = \xi/R$. For typical BEC parameters ($\xi \sim 0.3 \mu\text{m}$, $R \sim 50 \mu\text{m}$): $\varepsilon_{\text{blob}} \approx 0.006$, well within the point vortex regime. The blob correction (Proposition 10) is negligible: $\lambda(0.006) \approx \lambda_0 + O(10^{-4})$.

Finite-temperature effects. Current BEC vortex experiments operate at finite temperature ($T/T_c \sim 0.3\text{--}0.7$), where a thermal cloud coexists with the condensate and exerts a dissipative drag on vortex motion [?]. This drag damps the oscillation modes on a

timescale $\tau_{\text{drag}} \sim R^2/(D_v T/T_c)$ where D_v is the vortex diffusion coefficient. For the $N = 7$ quartic mode to be observable, the oscillation period (~ 100 s at $\varepsilon = 0.05$) must be shorter than τ_{drag} ; at $T/T_c = 0.5$ this requires $D_v \lesssim 10^{-9} \text{ m}^2 \text{ s}^{-1}$, which is achievable in large-radius traps ($R \geq 50 \text{ } \mu\text{m}$) at low temperature. The $N = 8$ stability prediction (unstable alone, stable with $q_0 = 1$) is less sensitive to dissipation because it concerns exponential growth vs. decay, not oscillation frequency; finite temperature adds a damping channel but does not change the sign of the eigenvalue.

8 Discussion and limitations

8.1 Why 2D and not 3D

The argument depends critically on two-dimensionality. The 2D Green’s function $G(r) = -\ln r/(2\pi)$ is the unique dilation-invariant pairwise interaction with $h' < 0$ (Remark 2; Corollary 3 lifts this to the N -body level). In 3D, the Green’s function $G(r) = -1/(4\pi r)$ has $G'(r) = 1/(4\pi r^2) > 0$ —the sign rule is violated, the N -gon is not an energy extremum along the spiral, and the stability analysis gives different results. The present mechanism—sign rule plus energy-Casimir stability—does not operate in 3D; whether other mechanisms can produce persistent polygonal vortex structures in three dimensions is a separate question.

8.2 What is novel

The individual components—Onsager condensation, Kraichnan cascades, Thomson stability, 2D Green’s functions—are established.

Important attribution note. The Havelock eigenvalue formula $\lambda_m = (N-1)-m(N-m)/2$ and the linear stability boundary $N \leq 7$ appear in Havelock [1931], §3, and have been known to the classical vortex dynamics community for nearly a century. Theorem 5 is therefore *not* new to hydrodynamicists. Its contribution is to the *statistical mechanics community*, where the polygon has been treated as an energy maximum selected by Onsager negative temperature [Onsager, 1949, Bouchet and Venaille, 2012]: the theorem demonstrates that the polygon is an energy *minimum* on the conserved-quantity surface, which is the opposite of the common assumption and changes the statistical interpretation substantially. Specialists in classical vortex dynamics who encounter Theorem 3 should read it as a bridge result, translating a known stability fact into the language of energy-Casimir minima.

What is new is: (1) identifying the N -gon as a *constrained energy minimum* (not maximum) on the conserved surface and deriving this from the Lagrange-multiplier structure (new to the Onsager literature; classical to Thomson–Havelock); (2) the analytic sign rule (Theorem 1) proving $H''(0) < 0$ for all interactions with $h' < 0$ and $(rh')' \leq 0$; (3) the three-scale separation showing the Onsager argument applies to cluster *formation* while the polygon *arrangement* is deterministic stability; (4) the complementary selection framework unifying Saturn (Rossby) and Jupiter (Thomson); (5) the observation that the logarithmic interaction’s borderline status in the sign rule makes it the “least robust” polygon producer—any other qualifying interaction gives stronger selection; (6) the spherical stability boundary $\varphi_{\text{crit}}(N)$ showing the sphere *destabilizes* relative to the plane (the Lagrange shift vanishes at the equator), with Saturn’s hexagon inside the stable zone,

and the exact rational thresholds $\xi_{\text{crit}}(N) \in \mathbb{Q}$ (equation (19)) with the geometric identity $\cos \varphi_{\text{crit}} = m(N-m)/(2(N-1))$; (7) the K_0 suppression transition with two distinct phase boundaries (spiral and constrained) and the class (iii) interaction landscape; (8) the vortex blob bridge showing regularization preserves and enhances stability, resolving the $N = 7$ marginal case, with profile independence via a Fourier dominance argument (Section 5); (9) the algebraic field extension pattern $\xi^*(N) \in \mathbb{Q}(\sqrt{\text{sq_free}(N-1)})$ for even N and $\mathbb{Q}(\sqrt{\text{sq_free}(N-3)})$ for odd $N \geq 9$, with $N = 10$ as the unique rational threshold $\xi^*(10) = 1/7$ (Section 4.3, equation (25)); and (10) the circulation disorder resilience analysis showing that the N_{crit} boundary is a stability domain rather than a knife-edge, with the $N = 7$ marginal ring identified as the unique disorder-sensitive configuration (Section 5.4.4).

Relation to prior work. The connection between Onsager statistics and large-scale vortex formation is developed in Bouchet and Venaille [2012] and Chavanis and Sommeria [1996], who classify self-organized vortex states in bounded domains using statistical mechanical methods. Aref et al. [2003] reviews vortex crystal configurations and their stability. The present contribution differs in three respects: (i) the closed-form formula for $H''(0)$ (equation (6)) does not appear in these reviews, which analyze stability through eigenvalues of the dynamical matrix rather than energy curvature along a specific deformation (the Thomson–Havelock literature has this eigenvalue result, but not via the energy-curvature path); (ii) the analytic sign rule (Theorem 1) identifying the precise conditions on h that guarantee $H''(0) < 0$ is new; and (iii) the constrained Hessian result (Theorem 5) corrects a widespread assumption *in the statistical mechanics literature* that polygonal configurations are energy maxima selected by negative temperature [Bouchet and Venaille, 2012, Chavanis and Sommeria, 1996]; this result is not new to the classical vortex dynamics literature [Thomson, 1883, Havelock, 1931], where the stability boundary is well established.

8.3 Limitations and scope

The mathematical results—the constrained energy minimum (Theorem 5), the sign rule (Theorem 1), and the stability table—are proven for discrete point vortices. They apply directly to Jupiter’s polar cyclone rings. For Saturn’s hexagon, which is a continuous jet meander, the relevant mechanism is Rossby wave stationarity of the QGPV equation, not the point vortex energy landscape. The continuous and discrete descriptions share the logarithmic Green’s function; the vortex blob analysis (Section 5.4) establishes that the stability boundary is preserved under regularization and that the $N = 7$ marginal case is resolved.

The Saturn bridge (Section 5.4.3) connects the continuous QGPV eigenvalue problem to the point vortex Hessian via critical-layer concentration. The key mechanism is $\mathbb{Z}_N$ symmetry: for any $\mathbb{Z}_N$ -symmetric cat’s-eye structure, the lobe circulations are equal and the centroids lie on the N -gon vertices, so the constrained Hessian has the same inertia as the Havelock Hessian (Remark 7). This is verified numerically for both smooth (Stuart vortex) and step-PV (Prandtl–Batchelor) cat’s-eye models. The braid vorticity fraction vanishes as $O(1-\eta)$ in the Stuart concentration limit (Remark 7), satisfying the conditions of Proposition 11.

What remains unproven is the *quantitative* bridge at Saturn’s actual parameters: the hexagonal jet has finite width $\sigma/R \approx 0.15$, not $\sigma/R \rightarrow 0$, and Saturn’s long-lived cat’s-eye is in the Prandtl–Batchelor regime (homogenized PV inside the lobes) where the

separatrix carries a thin vorticity sheet whose effect on the Hessian eigenvalues is bounded but not rigorously estimated. A definitive closure would require either resolving this sheet contribution or showing it is absorbed by the existing spectral gap $\lambda_{\min}(6) = 1/2$.

The Saturn hexagon is therefore supported by three arguments: (i) Rossby stationarity (Proposition 8) selects $n = 6$ independently of the energy landscape; (ii) geometric consistency with the spherical stability boundary (Section 4.2); and (iii) the critical-layer bridge (Remark 7), which establishes Hessian inertia preservation for $\mathbb{Z}_N$ -symmetric cat's-eye structures in the concentration limit. The first two arguments are independent of the bridge; the third provides the unifying connection to the point vortex energy landscape.

The Onsager negative-temperature hypothesis is a physical assumption whose applicability to Jupiter's polar region is not directly tested here. It is the key step that distinguishes $N = 8$ from $N = 7$ in the selection argument (Proposition 9): without negative temperature, the framework identifies $N = 8$ as *stable* but does not uniquely select it over $N = 7$.

The Siegelman et al. [2022] simulations provide the most direct evidence available. Their 2D turbulence on polar caps develops an inverse energy cascade that drives kinetic energy to the largest scales consistent with the polar geometry—the signature of a Onsager negative-temperature state. The polygon crystal is the maximum- H configuration within the Thomson-stable set: starting from random initial conditions, the simulations converge to the polygon ring with a central cyclone, which is the $N_{\max}$ configuration under Proposition 9 (they observe $N = 8$ north and $N = 5$ south, matching the Juno observations; the south-pole discrepancy from the Thomson/Onsager prediction is attributed to geometric packing as discussed above). This is indirect but consistent evidence that the planetary system is near negative temperature. A direct test would require measuring the energy spectrum at the polar scale in Juno or a dedicated mission.

Cassini and Juno data are digitized from published figures.

Dynamical maintenance. The constrained energy minimum (Theorem 5) is a static result: it identifies the polygon as the lowest-energy arrangement on the conserved-quantity surface, and by Arnold's energy-Casimir stability theorem [Arnold, 1969] this minimum is Lyapunov stable under the conservative 2D Euler dynamics—perturbations stay bounded. Whether this translates into an attractor under convective forcing depends on the ratio $\mathcal{R} = \tau_{\text{inject}}/\tau_{\text{restore}}$.

For Jupiter's $N = 8$ ring (Juno parameters: $R_{\text{ring}} \approx 7,500$ km, $\kappa_0/\kappa \approx 1$; Adriani et al. 2018), we use the full-cyclone circulation $\kappa = 2\pi r_c V_{\max} \approx 1.6 \times 10^9 \text{ m}^2 \text{ s}^{-1}$ with $r_c \approx 3,200$ km (the full Juno-observed cyclone radius) and $V_{\max} \approx 80 \text{ m s}^{-1}$. This is the appropriate measure of total vortex strength for computing orbital periods and eigenfrequencies. It differs from the Rankine-core estimate $\kappa_{\text{Rank}} \approx 3.5 \times 10^8 \text{ m}^2 \text{ s}^{-1}$ (using only $r_c \approx 700$ km, the solid-body rotation region) that enters the regime number $\beta R^3/\kappa$ in Section 5: the regime criterion requires only a characteristic vorticity scale, for which the core is sufficient.

- *Restoring:* The ring orbital period is $T_{\text{orb}} = 8\pi^2 R^2/[\kappa(N-1)] \approx 4.6$ days. The most marginal mode ($m = 4$) has Lagrangian eigenvalue $\lambda_{\text{eff}} \approx 3$ (bare eigenvalue -1 , raised to $\approx +3$ by the central cyclone with $\kappa_0/\kappa = 1$). The oscillation e-folding time is $\tau_{\text{restore}} = (\sqrt{\lambda_{\text{eff}}} \Omega)^{-1} \approx 0.4$ days.
- *Injection:* The upper-bound convective power into one cyclone's footprint is $P = F_{\text{int}} \pi r_c^2$, where $F_{\text{int}} \approx 7.5 \text{ W m}^{-2}$ is Jupiter's interior heat flux [Guillot et al., 2004]; with $r_c = 700$ km, $P \approx 1.15 \times 10^{13} \text{ W}$. The tropospheric kinetic energy per

cyclone is estimated as $E_K \approx \frac{1}{2}\rho V_{\max}^2 \pi r_c^2 H$ with $\rho \approx 0.2 \text{ kg m}^{-3}$, $H \approx 40 \text{ km}$, $V_{\max} \approx 80 \text{ m s}^{-1}$: $E_K \approx 4 \times 10^{19} \text{ J}$, giving $\tau_{\text{inject}} = E_K/P \approx 40 \text{ days}$.

The ratio $\mathcal{R} \approx 40/0.4 \approx 100$: the restoring force operates $\sim 100\times$ faster than the upper-bound convective injection. Including the inverse-cascade efficiency (typically $\lesssim 1\%$ of the heat flux reaches cyclone scales) raises $\mathcal{R}$ to order 10^4 . The Juno four-year observational baseline provides a direct empirical lower bound: $\mathcal{R} \gtrsim 4 \text{ yr}/0.4 \text{ days} \approx 3500$. The polygon is firmly in the restoring-dominated regime; convective forcing is a persistent perturbation that the energy minimum absorbs, not a mechanism that competes with the minimum for selection of the configuration.

Moist convection as vorticity source. The leading mechanism proposed for maintaining Jupiter’s polar cyclones in the observational literature [O’Neill et al., 2015] is moist convective plumes injecting cyclonic vorticity preferentially at high latitudes. The geometric argument is that near the pole, the β -effect preferentially suppresses anticyclonic plumes (their vorticity anomaly is opposed by the planetary background), while cyclonic plumes are unimpeded; the result is a net positive-vorticity injection into the polar cap. This mechanism is quite different from—and more specific than—the Onsager inverse-cascade picture invoked here.

The two mechanisms are complementary rather than competing, operating at different scales and addressing different aspects of the problem:

- **O’Neill et al. generation:** moist convection injects cyclonic PV into the polar region, selecting *sign* (cyclones, not anticyclones). This is the vorticity *source*.
- **Onsager organization:** the injected cyclonic PV undergoes an inverse energy cascade, clustering into N coherent vortices. This is scale-blind to the source mechanism.
- **Thomson/energy landscape:** the N clusters arrange into a regular polygon. This is determined entirely by the point-vortex energy function.

The Onsager inverse cascade (and the present paper) requires only that same-sign vorticity be present in the polar region; it does not predict the sign. The O’Neill mechanism provides that sign selection. Without it, the Onsager picture would predict equal probability of cyclonic and anticyclonic polar clusters—which is not observed. A complete theory of Jupiter’s polar polygons therefore requires both mechanisms: O’Neill for the vorticity source and polarity, and the present Thomson/Onsager framework for the organization and stability of whatever is produced.

Three configurations from two planets is consistent with the proposed mechanism but not sufficient to establish universality. Uranus and Neptune observations would substantially strengthen the case. We use “consistent with a general mechanism” rather than “universal” to reflect the available evidence.

A The Havelock identity

Lemma 16. *For integers $N \geq 2$ and $1 \leq m \leq N - 1$,*

$$T_m := \sum_{p=1}^{N-1} \frac{1 - \cos(2\pi pm/N)}{4 \sin^2(\pi p/N)} = \frac{m(N-m)}{2}.$$

This identity appears in Havelock [1931], §3, as a step in computing the stability eigenvalues of the N -vortex ring; see also Aref et al. [2003], Appendix A.

Proof. Write $\omega = e^{2\pi i/N}$. Using $2 \sin(\pi p/N) = |1 - \omega^p|$, expand $[1 - \cos(2\pi pm/N)]/[4 \sin^2(\pi p/N)] = \frac{1}{2} |\sum_{k=0}^{m-1} \omega^{pk}|^2 = \frac{1}{2} \sum_{j=0}^{m-1} \sum_{k=0}^{m-1} \omega^{p(j-k)}$. Summing over $p = 1, \dots, N-1$ and using the orthogonality relation $\sum_{p=1}^{N-1} \omega^{p\ell} = N\delta_{\ell \equiv 0 \pmod{N}} - 1$:

$$T_m = \frac{1}{2} \sum_{j,k=0}^{m-1} (N\delta_{j \equiv k \pmod{N}} - 1) = \frac{1}{2} (Nm - m^2) = \frac{m(N-m)}{2},$$

where $\delta_{j \equiv k}$ contributes m (since $0 \leq j, k \leq m-1 < N$ forces $j = k$) and the -1 contributes $-m^2$. $\square$

B Derivation of $C_1(\mathbf{H}^2, \xi)$ and the near- $K = 0$ expansion

We work in the Poincaré disk of radius a and set $a = 1$ without loss of generality (the formula is dimensionless in $\xi = r_E^2/a^2$).

$\mathbf{H}^2$ Hamiltonian and moment map. In Poincaré disk coordinates the N -vortex Hamiltonian is [Kimura, 1999]

$$H_{\text{hyp}} = -\sum_{j < k} \ln |z_j - z_k| + \frac{N-1}{2} \sum_k \ln(1 - |z_k|^2), \quad (40)$$

equivalent to $-\sum_{j < k} \ln \sinh(d_{\mathbf{H}^2}(j, k)/2)$ up to an additive constant, via the identity $\sinh(d/2) = |z_j - z_k|/\sqrt{(1 - |z_j|^2)(1 - |z_k|^2)}$. The $\text{SO}(2, 1)$ angular moment map is $J = \sum_k (1 + |z_k|^2)/(1 - |z_k|^2)$.

Equilibrium and rotation rate Ω . The N -gon at Euclidean radius r_E ($\xi = r_E^2$) satisfies the Lagrangian stationarity condition $\partial H_{\text{hyp}}/\partial x_0 = \Omega \partial J/\partial x_0$ at vortex $0 = (r_E, 0)$:

$$\begin{aligned} \frac{\partial H_{\text{hyp}}}{\partial x_0} &= -\frac{N-1}{2r_E} - \frac{(N-1)r_E}{1-\xi} = -\frac{(N-1)(1+\xi)}{2\sqrt{\xi}(1-\xi)}, \\ \frac{\partial J}{\partial x_0} &= \frac{4r_E}{(1-\xi)^2} = \frac{4\sqrt{\xi}}{(1-\xi)^2}. \end{aligned}$$

Dividing gives

$$\Omega = -\frac{(N-1)(1+\xi)(1-\xi)}{8\xi}. \quad (41)$$

Mode- m second derivatives. Let $\hat{u}$ be the normalized radial Fourier mode of wavenumber m ($\|\hat{u}\| = 1$, Euclidean). Two ring-geometry identities hold for all $1 \leq m \leq \lfloor N/2 \rfloor$:

$$\sum_k |\hat{u}_k|^2 = 1, \quad \sum_k [\text{Re}(z_k \overline{\hat{u}_k})]^2 = r_E^2 = \xi.$$

The three contributions to $\lambda_m \cdot r_E^2$ are:

(1) *Flat-plane Havelock term*. By scale invariance of $-\ln|\cdot|$, $\partial^2 H_{\text{flat}}/\partial \varepsilon^2 = \sigma_m/r_E^2$ with $\sigma_m = (N-1)/2 - m(N-m)/2$. Multiplying by r_E^2 :

$$(1) = \frac{N-1}{2} - \frac{m(N-m)}{2}.$$

(2) *Curvature-correction term* W . With $W = \frac{N-1}{2} \sum_k \ln(1 - |z_k|^2)$, expanding at $\varepsilon = 0$:

$$\frac{\partial^2 W}{\partial \varepsilon^2} = \frac{N-1}{2} \sum_k \left[\frac{-2|\hat{u}_k|^2}{1-\xi} - \frac{4[\text{Re}(z_k \overline{\hat{u}_k})]^2}{(1-\xi)^2} \right] = -\frac{(N-1)(1+\xi)}{(1-\xi)^2}.$$

Multiplying by $r_E^2 = \xi$:

$$(2) = -\frac{(N-1)\xi(1+\xi)}{(1-\xi)^2}.$$

(3) *Lagrange-multiplier term* $-\Omega J$. Expanding J similarly gives

$$\frac{\partial^2 J}{\partial \varepsilon^2} = \frac{4}{(1-\xi)^2} \left[1 + \frac{4\xi}{1-\xi} \right] = \frac{4(1+3\xi)}{(1-\xi)^3}.$$

With $-\Omega = (N-1)(1-\xi^2)/(8\xi)$ from (41):

$$-\Omega \frac{\partial^2 J}{\partial \varepsilon^2} = \frac{(N-1)(1+\xi)(1+3\xi)}{2\xi(1-\xi)^2}.$$

Multiplying by $r_E^2 = \xi$:

$$(3) = \frac{(N-1)(1+\xi)(1+3\xi)}{2(1-\xi)^2}.$$

Algebraic simplification. The ξ -dependent part of (2) + (3) (factor $N-1$):

$$-\frac{\xi(1+\xi)}{(1-\xi)^2} + \frac{(1+\xi)(1+3\xi)}{2(1-\xi)^2} = \frac{(1+\xi)}{2(1-\xi)^2} [(1+3\xi) - 2\xi] = \frac{(1+\xi)^2}{2(1-\xi)^2}.$$

Adding term (1):

$$C_1 = \frac{N-1}{2} + \frac{(N-1)(1+\xi)^2}{2(1-\xi)^2} = \frac{N-1}{2} \cdot \frac{(1-\xi)^2 + (1+\xi)^2}{(1-\xi)^2} = \frac{(N-1)(1+\xi^2)}{(1-\xi)^2},$$

establishing (20).

Near- $K = 0$ expansion. For $N = 7$, $m = 3$: $m(N-m)/2 = 6 = N-1$. Therefore

$$\lambda_3 \cdot r_E^2 = C_1 - 6 = 6 \cdot \frac{(1+\xi^2) - (1-\xi)^2}{(1-\xi)^2} = \frac{12\xi}{(1-\xi)^2}.$$

Using $\xi = r_E^2/a^2 = |K|r_E^2$:

$$\lambda_3 = \frac{12|K|}{(1-\xi)^2} = 12|K| + O(K^2 r_E^2).$$

The coefficient $12 = 2m(N-m)/2|_{m=3, N=7}$ arises from the cancellation $(1+\xi^2) - (1-\xi)^2 = 2\xi$, which holds because $N-1 = m(N-m)/2$ exactly at this critical mode.

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